Sunway Diploma Studies

IT Project

Smart Math

by

Koh Pei Yang

Chin Kai Yang

Diploma in Information Technology

Supervisor: **Ms. Seetha Letchumy M.Belaidan**

Mr Kevin Foo Kai Pin

28 November 2025

Project Title : Smart Math
Student : Koh Pei Yang, Chin Kai Yang
Supervisor : Ms.Seetha Letchumy M.Belaidan, Mr Kevin Foo Kai Pin

Abstract

Smart Math is a mobile application which developed for SPM students (Form 4 and Form 5). During the fact finding, we founded that student always struggle in Problem Solving-based question, which student can't understand question requirement or their calculation error. To solve these problems, Smart Math integrated chapter-based question, AI tutor chat-based system, video learning and scan image system in the application, which is helping student to practice mathematics question, improve their mathematics skills and learn the steps of solution while doing the practice. This system also supports a local SQLite database which can stores the local questions, local answer and student's answer. It's help student to review their answer and the steps of solutions that provided by AI tutor. To ensure the AI Tutor's answer is correct, this system will compare AI answer and database answer (local answer), using the calculate similarity method with 'AI answer to database answer', make sure that the correctness of AI answers. If the AI answers is incorrect, students also can review the local answer to see the correct answer.

Table of Contents

Abstract ..... **ii**

INTRODUCTION **1**

1.1 Introduction 1

1.2 Background 1

PROBLEM STATEMENT AND LITERATURE REVIEW **3**

2.1 Main Problem 3

2.2 Existing Work 4

PROJECT OBJECTIVES **6**

3.1 Project Objectives 6

PROPOSED SOLUTION **7**

4.1 System roles 7

4.2 Project Scopes/Model 7

4.3 Technology Involved 8

Project Plan **9**

5.1 Development Methodology 9

5.2 Conclusion of Methodology 10

5.3 Proposed Work Plan 11

Requirements **15**

6.1 Fact Findings 15

6.1.1 Reason of Fact Findings 15

6.1.2 Survey Result 15

6.2 Functional Requirements 24

6.3 Non-functional Requirements 25

6.4 Modelling 26
6.4.1 Use Case Diagram 26
6.4.2 Use Case Description 26
System Design 31
7.1 Architectural Design 31
7.2 Database Design 33
7.2.1 Entity Relationships Diagram (ERD) 33
7.2.2 Relational Schema 33
7.2.3 Data Definition Language 34
7.2.4 Data Dictionary 35
7.3 Prototype Design 36
System Deployment 48
8.1 Selected AI Chatbot 48
Discussion and Results 49
9.1 Achievements 49
9.4 Future Enchancements 50
References 52
Appendix 54
10.1 Progress Form 54
10.2 Survey Question 59

List of Figures and Tables

INTRODUCTION

1.1 Introduction

This project focuses on supporting Form 4 and Form 5 students in Malaysia which is preparing SPM Mathematics examination. As a SPM graduated students (2020,2021), we always face the challenges in understanding question concepts, identifying mistake on calculation and receiving immediate feedback during the examination. Students always faced challenge in understanding question concepts, identify mistake and receiving immediate feedback is because they always lack sufficient practice and clear explanations. To solve these problems, we developed a mobile application that helps students to prepare their SPM examination.

1.2 Background

According to the record in 2024 SPM Mathematics examination, 77.6% of SPM students “PASS” and 22.4% “FAIL” in SPM examination. These result show that Malaysian student just meets the minimum expected performance in this subject, they still need to improve and put more effort in there (Utusan Malaysia, 2024).

Mathematics is an important subject for Form 4 & 5 students in Malaysia. This is because, Universities / Collage in Malaysia usually require students to have a “CREDIT” in SPM Mathematics for admission, such as Diploma in IT, Accounting and some mathematics-related programs (Lembaga Peperiksaan Malaysia, 2021). A “CREDIT” in Mathematics can show the student’s logical thinking skill while practice the question. However, students often face the challenges of learning Mathematics because they find it difficult to understand concepts, identify mistakes in their steps and lack of practice.

Traditional exercise books are not useful for students. This is because most students will take a sneak peek at the exercise book’s answers, which will reduce student’s problem-solving skills. However, exercise books answer sheet will only contain the final answer of the question, meaning students will not be able to learn the steps of solving the problem, resulting in losing “process” mark during the SPM examination.

Online resources such as YouTube and digital practice platform also help students to improve their general Mathematics skills. However, they lack personalised explanations or detailed guide on how to solve specific problems. This makes it difficult for learners to fully understand the logic, strategy, and mathematical concepts required to solve problems independently.

Nowadays, many students always using AI tools such as ChatGPT and DeepSeek to support their studies. These tools can provide a personalized explanations for the

question such as Basic Linear Algebra, Calculus and Discrete Mathematics. In this way, students can learn the process to solve the question and ask follow up questions if they couldn’t comprehend the explanation.

PROBLEM STATEMENT AND LITERATURE REVIEW

2.1 Main Problem

SPM students commonly struggle with application-based Mathematics questions, particularly those involving formula interpretation, conceptual understanding, and multi-step calculations. Many students are unable to solve these questions confidently because they lack sufficient practice and do not receive targeted exposure to real exam-style problems. As a result, they often lose marks due to missing calculation steps, misunderstanding question requirements, or becoming confused midway through the solution process (Lim et al., 2021).

In addition, the current use of AI tools such as ChatGPT or DeepSeek also has limitation without generating AI answer or solution in picture. This is because the API limited picture generator for each user only can generate 6-7 pictures in one day (OpenAI, 2024). This becomes a challenge for students who want to practice many questions in a short period time, as they may not receive consistent guidance for multiple question during intensive revision periods. In the SPM Mathematics exam, questions are typically presented in three main formats:

Objective / Multiple-Choice Questions (MCQ):

These require quick reasoning but often involve traps or similar answer choices. Students struggle because they may know the formula but fail to apply it accurately under time pressure.

Subjective / Problem-Solving Questions:

These require multi-step calculations and clear working processes. Students often lose marks not because the final answer is wrong, but because they make mistakes in intermediate steps or fail to justify their reasoning.

Graph / Diagram-Based Questions:

These questions require interpretation of diagrams, tables, graphs, functions and visual data. Students find them challenging because they require both conceptual understanding and spatial interpretation, which are skills that many have not fully developed.

2.2 Existing Work

Some applications also try to solve the mathematic problems, such as Homework Helper and Mathphix. Here are the findings on both applications.

Homework Helper:

The Homework Helper serves as a AI assistant for all subjects, such as Mathematic, Chemistry, Physics, Biology, History, Geography and Literature. Its provide a camera scanning function for all the subjects, which user just scan the question and get the answer for ChatGPT, DeepSeek or Gemini. There will provide the guidance step-by-step to help user understand the concepts (Homework Helper & Math Solver, 2023).

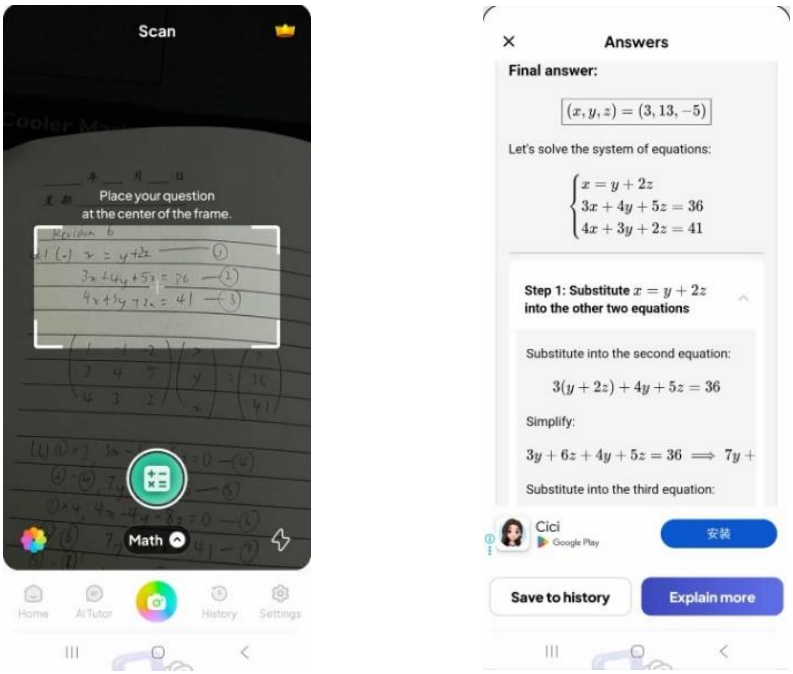


Figure 2.21 show the camera scan & AI assistant for Homework Helper

Mathphix:

The Mathphix is a mobile learning application designed to support secondary school students in learning Mathematics. The most difference between Mathphix and Homework Helper is the input method. Mathphix provide a hand-drawing function to ask the questions. It also provides a AI tutor for user checking answer which create more interactive and effective learning experience (Mathphix Snip, 2019).

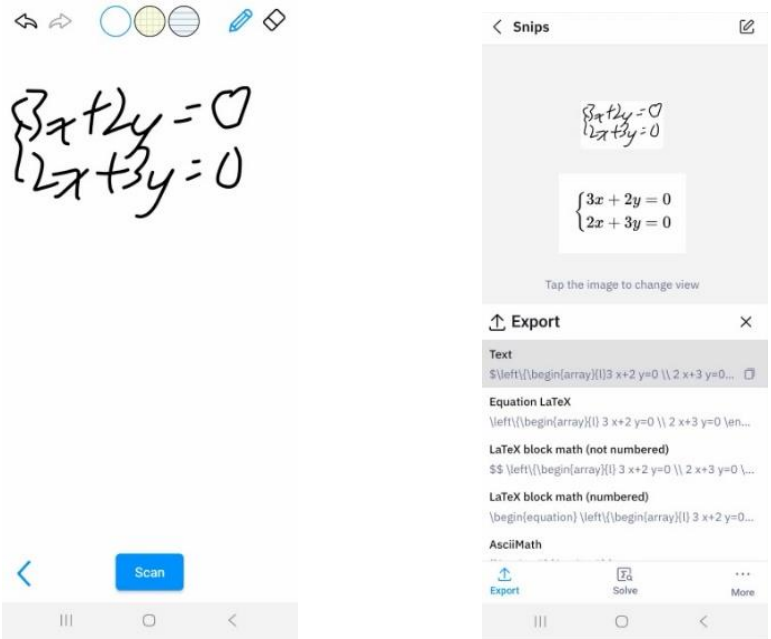


Figure 2.22 show the hand-drawing function in Mathphix

Weakness of both application:

Homework Helper & Mathphix are not suitable for Malaysia SPM syllabus, because their outline is against on IGCSE, SAT, IB, A-Level and US Common Core. Other than that, these kinds of AI applications don't provide an internal error-checking mechanism within their AI answer, which makes it difficult for students to identify and correct mistakes effectively. According to user feedback on Google Play Store, some users have reported inconsistency and limited accuracy in the generated answers, such as fail to provide correct answers for fixed-answer questions and the answers are different every time (Google Play Store, 2024).

PROJECT OBJECTIVES

3.1 Project Objectives

1. To develop Smart Math as a user-friendly mobile application, especially in solving mathematics word problems.
   - The application is designed to simplify the learning experience for SPM student. Smart Math offering a structured chapter-based question, which is follow the Mathematics SPM syllabus. The reason that Smart Math special focus on mathematics word problems is because many students struggle due to weak interpretation skills and lack of strategy. Smart Math aims to improve student's confidence and understanding in applying mathematical concepts to real-world scenarios (Lembaga Peperiksaan Malaysia, 2021).

2. To address student's difficulties in identifying mistakes and receiving immediate feedback by providing step-by-step AI guidance, comprehensive problem coverage, and AI-driven answer verification.
   - Many students fail in mathematics examination because they do not understand where the error from their calculation is. Smart Math addresses this gap by generating detailed reasoning steps, checking errors in student input, and offering corrective feedback instantly, which is helpful for student to understanding the question easily. The AI guidance covers subjective based question which providing students with continuous support that traditional textbooks or static answer sheets cannot offer (Herold, 2024).

3. To improve student learning outcome, we increased practice frequency and reducing common mistakes to measure this improvement through user done the assessment.
   - Consistent practice is a key factor in achieving excellence in mathematics. Smart Math encourages students to attempt more questions by offering on-demand feedback, accessible practice sets, and performance analytics. These features are helping student to train their memory retention through repeated practice.

PROPOSED SOLUTION

4.1 System roles

System roles define the responsibilities and functions of different participants within the Mathematic SPM Practice and AI guidance. Each role contributes to ensuring students can effectively practice questions, receive guidance and get a better result during the SPM examination.

i. **User (Students)**

   This project tends to focus on Form 4 & 5 students who are preparing the SPM examination. That is an extra exercise to help students to improve their Mathematica's skill especially word-problem based techniques. User no need to Log In or Sign Up the application, which means the process will be saved by local and just user can see the process. User can solve the mathematic problem, review AI-generated step-by-step guideline and making the correction with their answer.

4.2 Project Scopes/Model

In the context of educational applications, project scope and modules play a crucial role in defining the functionality and usability of the system. Each module contributes to a specific learning need, and together they form a complete platform for improving students' learning experience.

i. **Practice Module:**

   In Smart Math applications, the practical modules provide students with access to mathematics exercises based on the SPM syllabus. It covers Form 4(10 chapters) and Form 5(8 chapters), allowing students to choose specific chapters to practice. Each chapter contains 3-4 questions to strengthen understanding of key topics.

ii. **AI Guidance Module:**

   We use Gemini(v2.3) as our AI guidance module, which can help user to checking answer, provide step-by-step instructions and convenient for self-study.

4.3 Technology Involved

The tools that developed this application system.

Android Studio (IDE):

- Android Studio is the official integrated development environment (IDE) for Android app development. It provides comprehensive tools for designing coding and testing Android applications. The IDE supports Java and Kotlin languages, making it suitable for developing scalable apps (Meet Android Studio, n.d.).

Java (Programming Language):

- Java is a widely used, object-oriented programming language known for its portability and reliability. It has long been the foundation of Android app development, providing robust syntax and rich libraries (Meet Android Studio, n.d.).

SQLite (Database):

- SQLite is a lightweight, embedded relational database commonly used in mobile applications. In this project, it is integrated with Room, an abstraction layer provided by Android Jetpack, to simplify database access and provide compile-time SQL validation (SQLite, 2024; Android Developers, 2024).

OpenRouter (AI API Service):

- OpenRouter is a platform that provides access to various Large Language Models (LLMs) that support an API between server and user's devices. It enables integration of AI-powered features such as solving math problems, generating step-by-step explanations, and providing tutoring guidance (OpenRouter, 2024).

XML (UI/UX Design):

- XML (Extensible Markup Language) is the primary markup language used in Android development for designing Smart Math interface. Developer can design the layout using buttons, icons and images. Smart Math using the XML to build all front-end design.

Project Plan

5.1 Development Methodology

Rapid Application Development (RAD):

Rapid Application Development (RAD) is a software development methodology that allow a quick prototyping which allows user to give feedback quickly and modifying the prototype base on user needs. By having multiple iterative cycles, the developers can ensure the build matches user's expectations throughout the refinement of different modules in the system (Kissflow, 2024). RAD very is suitable in this proposed project because it enables developers to quickly test the AI guidance feature and improve question modules based on student and teacher feedback.

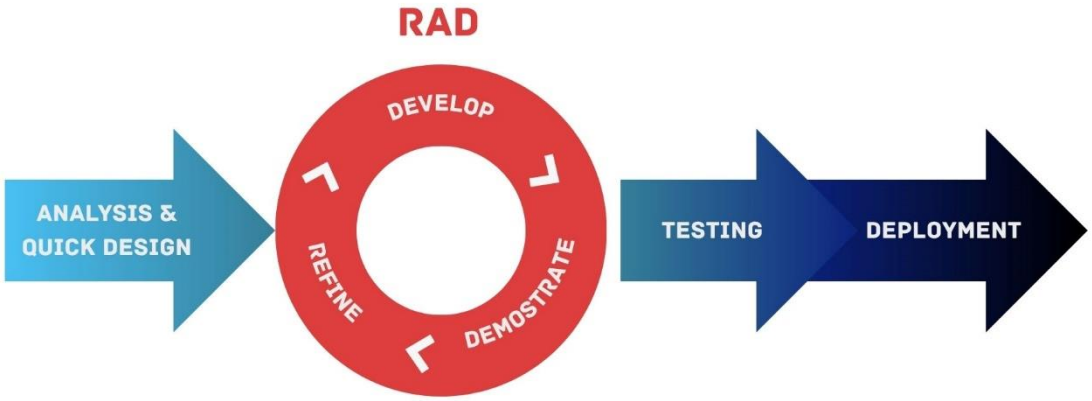


Figure 5.1: Rapid Application Development

1. **Analysis and Quick Design**

   This stage is focused on making a simple design while gathering user's requirement to make an analysis and get overview on how to refine the design. For example, we are required to have discussion with SPM students and teachers to scope key features, such as AI-driven problem solving, database integration, and user-friendly interface layout. The aim of this phase is to ensure the alignment of our system structure with the needs of SPM Mathematics candidates. Estimated time of completing this phase is around 3 weeks.

2. **Prototype Cycle (Develop → Demonstrate → Refine → Develop)**

   In this iterative cycle, the developers will start off by building a prototypes then they will demonstrate it to users to collect their feedbacks and based on the feedback they will continuously improve the prototype until it fulfil user's requirements. This phase in the context of this project is that it could help us improve accuracy of AI-generated explanations, enhance question display modules, and adjust features based on actual user experience. Estimated time of completing this phase is around 4 weeks.

3. **Testing**

   In this phase, the system can test the functionality in Smart Math, such as OCR, LaTeX and AI answers. This testing phase is ensured that AI guidance can work properly in the system. For example, during the testing, one of the benchmarks is that user can get the expected AI generated result. Estimated time of completing this phase is around 10 weeks.

4. **Deployment**

   After successful testing, the final version of the Smart Math system will be deployed for real student use. SPM students can used to asking AI explanation and practice the chapter-based questions. Estimated time of completing this phase is around 4 weeks.

5.2 Conclusion of Methodology

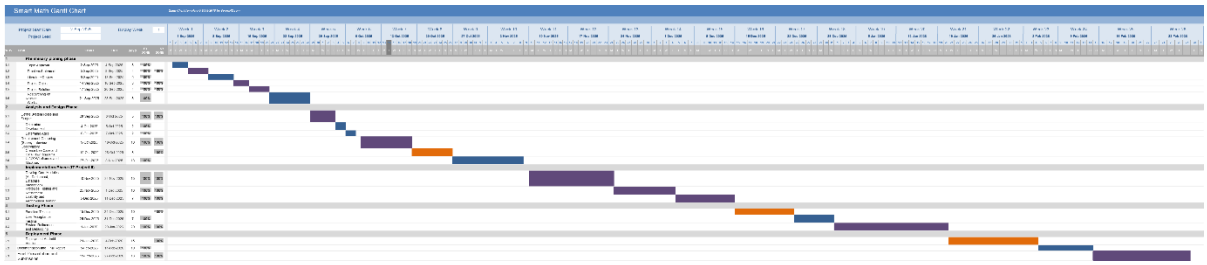
Aspect	**RAD (Rapid Application Development)**
Core Concept	Focuses on quick development and iterative prototyping to gather user feedback rapidly.
Time Structure	No fixed development cycle — development proceeds through phases and prototypes.
Team Roles	No specific roles are defined (developers and designers collaborate as needed).
Main Focus	User involvement and fast adjustments based on feedback.

5.3 Proposed Work Plan

No	Activities	Duration (days)	Start Time	End Time	Task Mode
1	**Preliminary planning phase**	**27**	**2 Sep 2025**	**28 Sep 2025**	
1.1	Topic Approval	3	2 Sep 2025	4 Sep 2025	Done
1.2	Problem Statement	4	5 Sep 2025	8 Sep 2025	Done
1.3	Literature Review	5	9 Sep 2025	13 Sep 2025	Done
1.4	Project Object	3	14 Sep 2025	16 Sep 2025	Done
1.5	Project Solution	4	17 Sep 2025	20 Sep 2025	Done
1.6	Researching on Related Works	8	21 Sep 2025	28 Sep 2025	Done
2	**Analysis and Design Phase**	**41**	**29 Sep 2025**	**8 Nov 2025**	
2.1	Define System Roles and Scopes	5	29 Sep 2025	3 Oct 2025	Done
2.2	Determine Development Methodology (RAD)	2	4 Oct 2025	5 Oct 2025	Done
2.3	Determine Agile Approach	2	6 Oct 2025	7 Oct 2025	Done
2.4	Requirement Gathering (Survey, Interview, Observation)	10	9 Oct 2025	18 Oct 2025	Done
2.5	Create Use Case and Data Flow Diagrams	8	19 Oct 2025	26 Oct 2025	Done
2.6	UI/UX Wireframes and Mockups	13	27 Oct 2025	8 Nov 2025	Done
3	**Implementation Phase (IT Project II)**	**40**	**10 Nov 2025**	**14 Dec 2025**	
3.1	Develop Core Modules (AI, Dashboard, Database Integration)	15	10 Nov 2025	24 Nov 2025	Done
3.2	Prototype Testing and Refinement	10	25 Nov 2025	4 Dec 2025	Done
3.3	Usability and Performance Testing	10	5 Dec 2025	14 Dec 2025	Done
4	**Testing Phase**	**37**	**15 Dec 2025**	**11 Jan 2026**	
4.1	Function Testing	10	15 Dec 2025	24 Dec 2025	Done
4.2	User Acceptance Testing	7	25 Dec 2025	31 Dec 2025	Done
4.3	System Refinement and Debugging	20	1 Jan 2026	20 Jan 2026	Done
5	**Deployment Phase**	**38**	**21 Jan 2026**	**27 Feb 2026**	

5.1	Deployment Android Version	15	21 Jan 2026	4 Feb 2026	Done
5.2	Documentation and Final Report	10	5 Feb 2026	14 Feb 2026	Done
5.3	Final Presentation and Submission	13	15 Feb 2026	27 Feb 2026	Done

Table 5.3: Task name for the work plan



Blue: Koh Pei Yang
Orange: Chin Kai Yang
Purple: Both

Chart 5.3: Gantt chart for work plan

Figure 5.3.1 Scrum Board for work plan

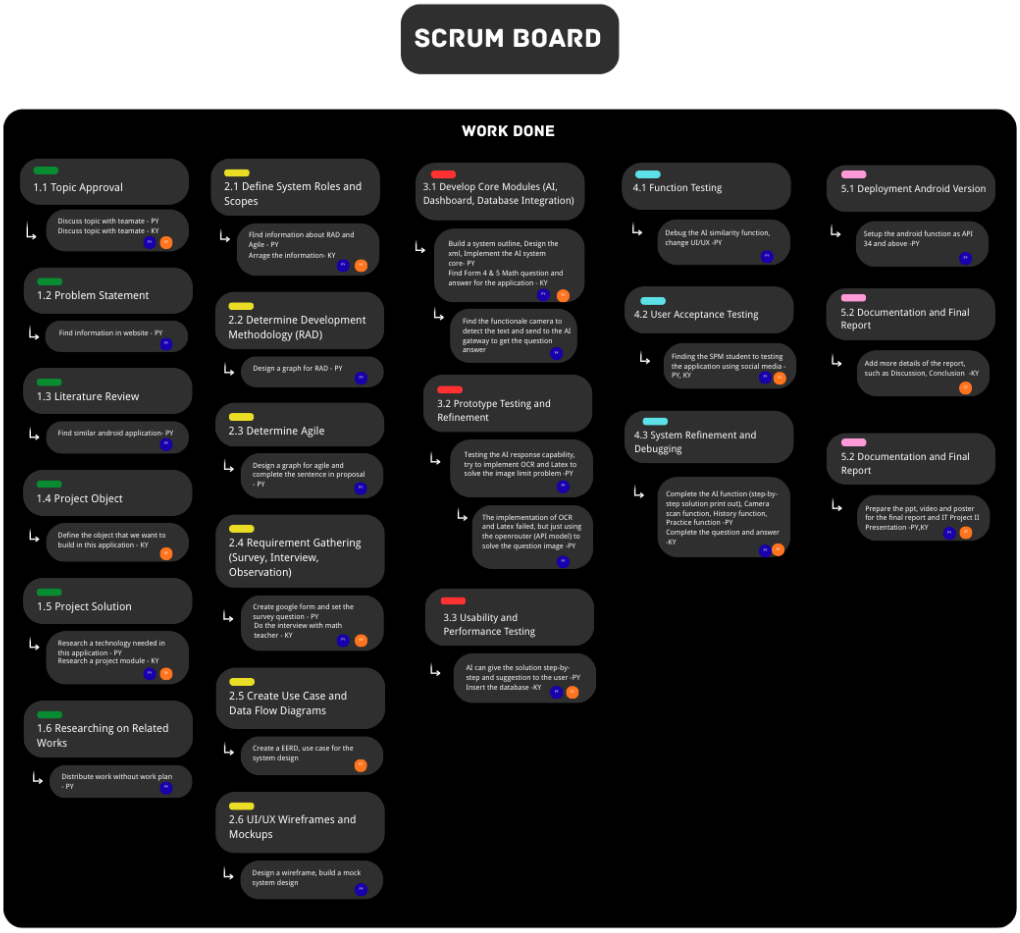


Figure 5.3.2 Scrum Board for work plan in IT Project (II)

Requirements

6.1 Fact Findings

6.1.1 Reason of Fact Findings

Fact Finding is very important for developing an application for student, because the main purpose is to show the value of this application, which can attract more students and users to use our applications. We conducted the survey by using Google Form as our fact finding method.

6.1.2 Survey Result

To support the fact findings presented above, a survey was conducted among secondary school students and teachers.

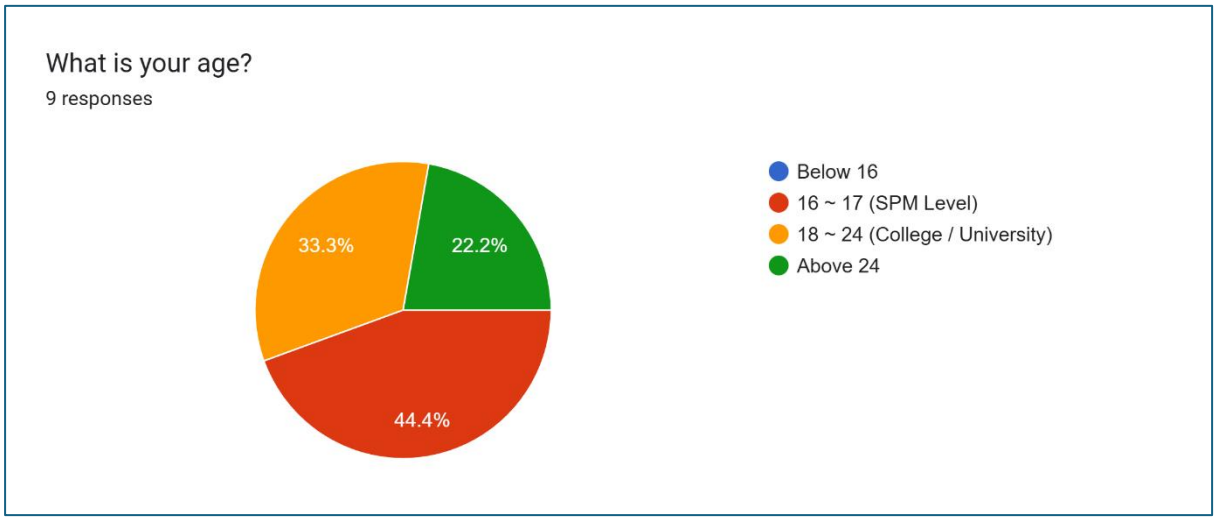


Figure 6.1.2.1: Pie Chart of Age Distribution

The feedback received from 9 respondents show a distribution ages. The highest percentage is 44.4% for 16-17 (SPM Level) years, 33.3% for 18-24 (College/University) and 22.2% for Above 24 years.

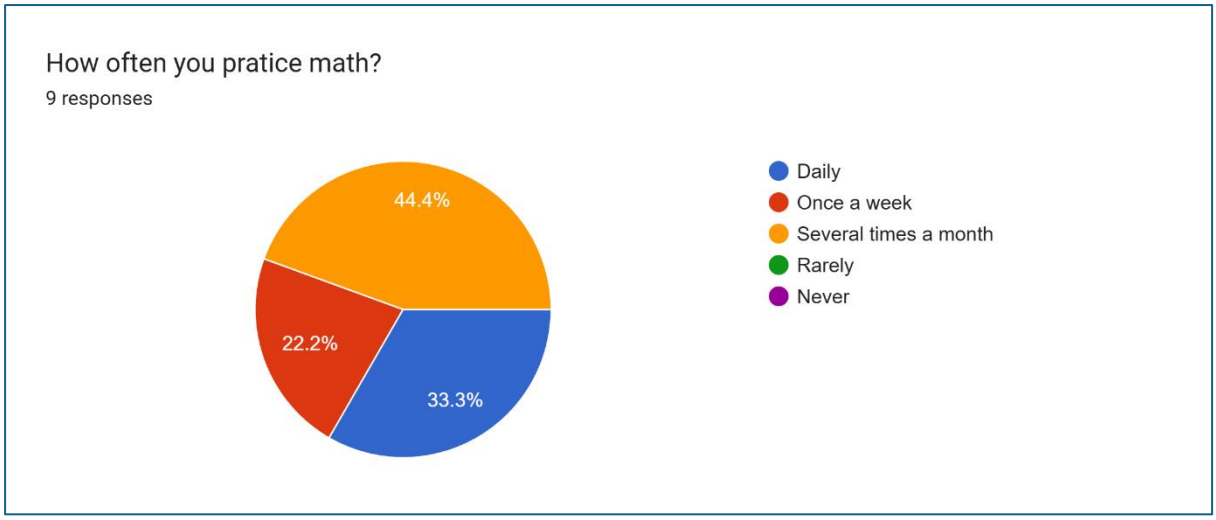


Figure 6.1.2.2: Pie Chart of Practice Frequency

This question was designed to understand the frequency of mathematics practice. The highest percentage is 44.4% for Several times a month, 33.3% for daily and 22.2% for once a week.

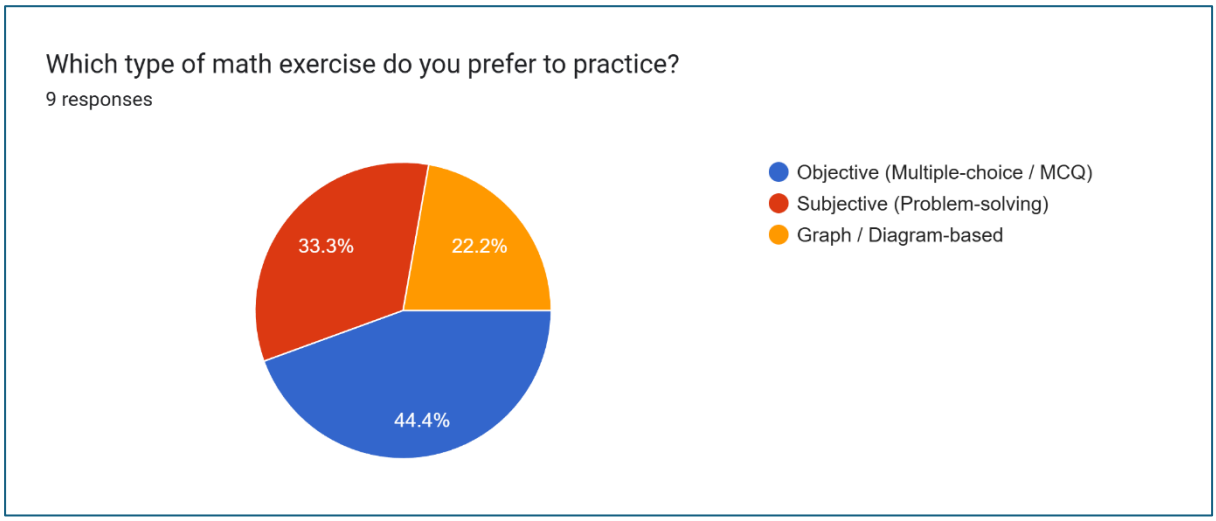


Figure 6.1.2.3: Pie chart of preferred math exercise types

The highest percentage is 44.4% for Objective (Multiple-choice/MCQ), 33.3% for Subjective (Problem-solving) and 22.2% for Graph/Diagram-based.

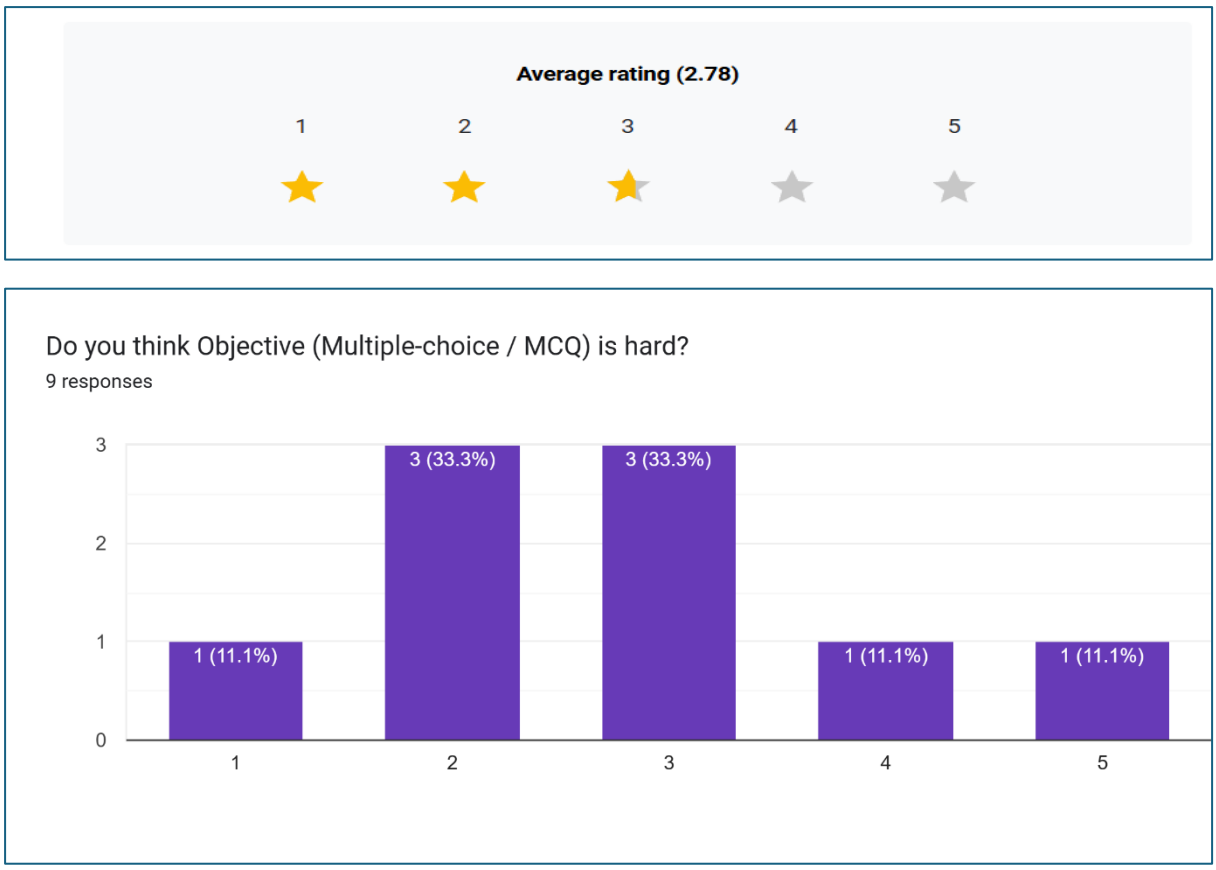


Figure 6.1.2.4: Rating Difficulty of MCQ

Figure 6.1.2.4 shows the difficulty ratings given by MCQ. The highest percentage is 33.3% for both 2-star and 3-star rating, 11.1% for 1-star, 4-star and 5-star. The overall average rating is 2.78, suggesting that most students do not find MCQ questions very difficult.

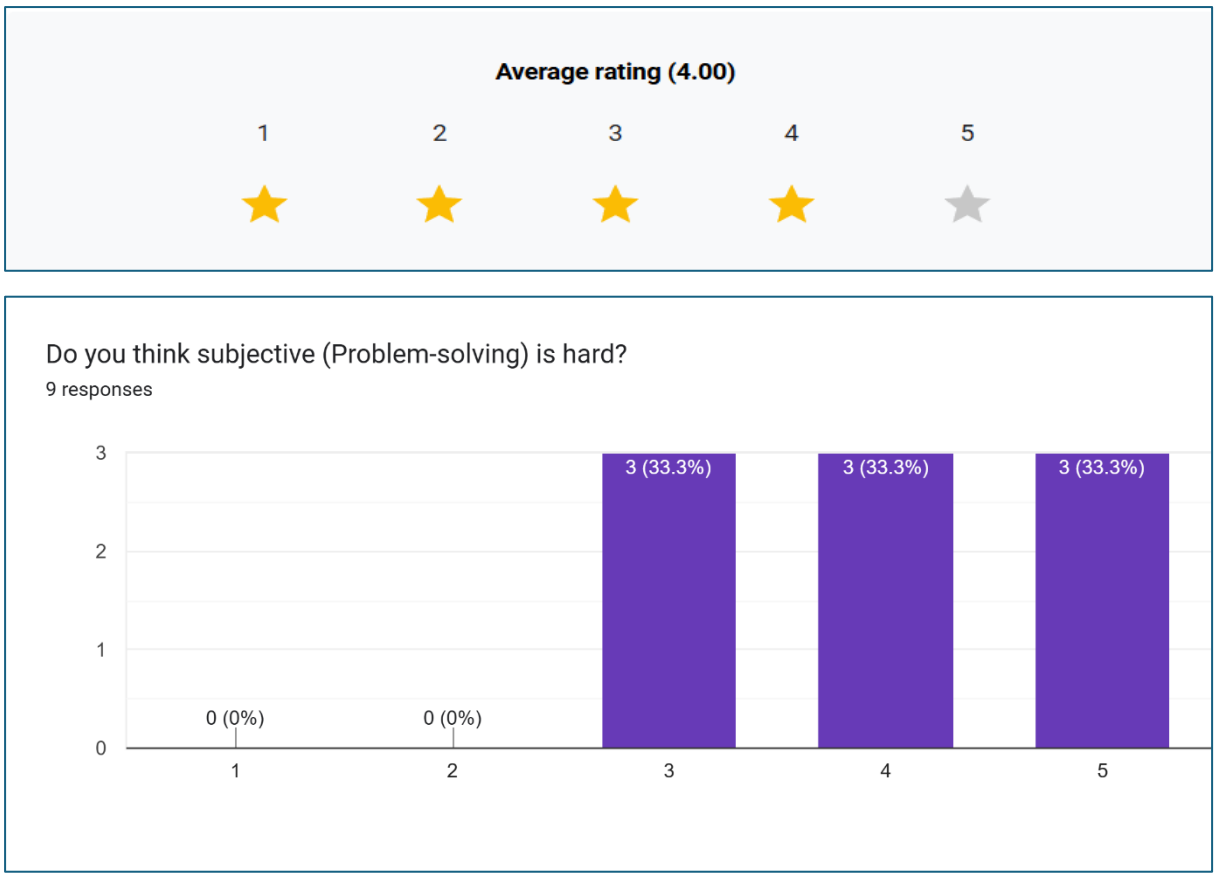


Figure 6.1.2.5: Rating Difficulty of Subjective

Figure 6.1.2.5 shows the difficulty ratings given by Subjective. The highest percentage is 33.3% for 3-star 4-star and 5-star rating. The overall average rating is 4, suggesting that most students find Subjective questions is very difficult.

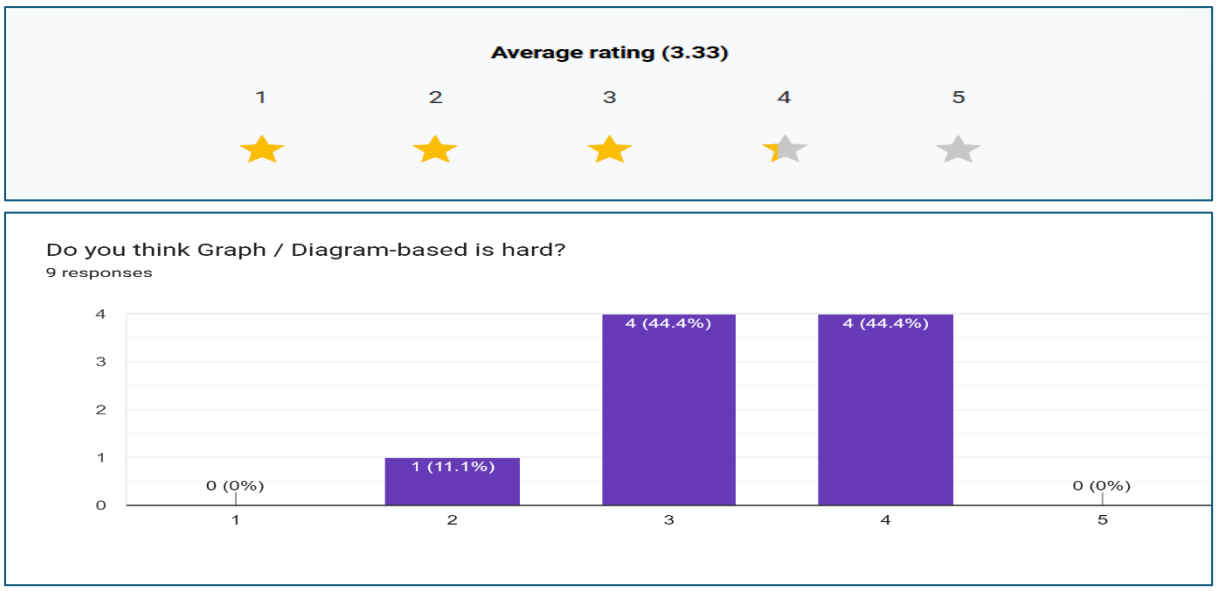


Figure 6.1.2.6: Rating Difficulty of Graph/Diagram-based

Figure 6.1.2.6 shows the difficulty ratings given by Graph/Diagram-based. The highest percentage is 44.4% for 3-star and 4-star rating, 11.1% for 2-star rating. The overall average rating is 3.33, suggesting that most students find Graph/Diagram-based questions is easier than Subjective questions.

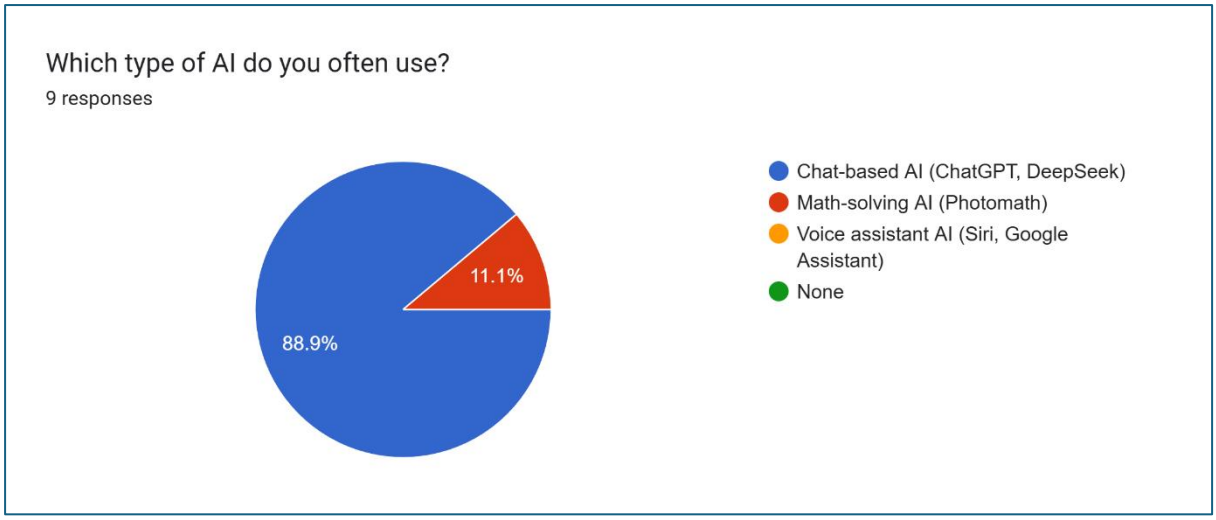


Figure 6.1.2.7: Pie chart of AI distribution

Figure 6.1.2.7 shows the types of AI tools commonly used by respondents. The highest percentage is 88.9% Chat-based AI (ChatGPT, DeepSeek), 11.1% for Math-Solving AI (Photomath).

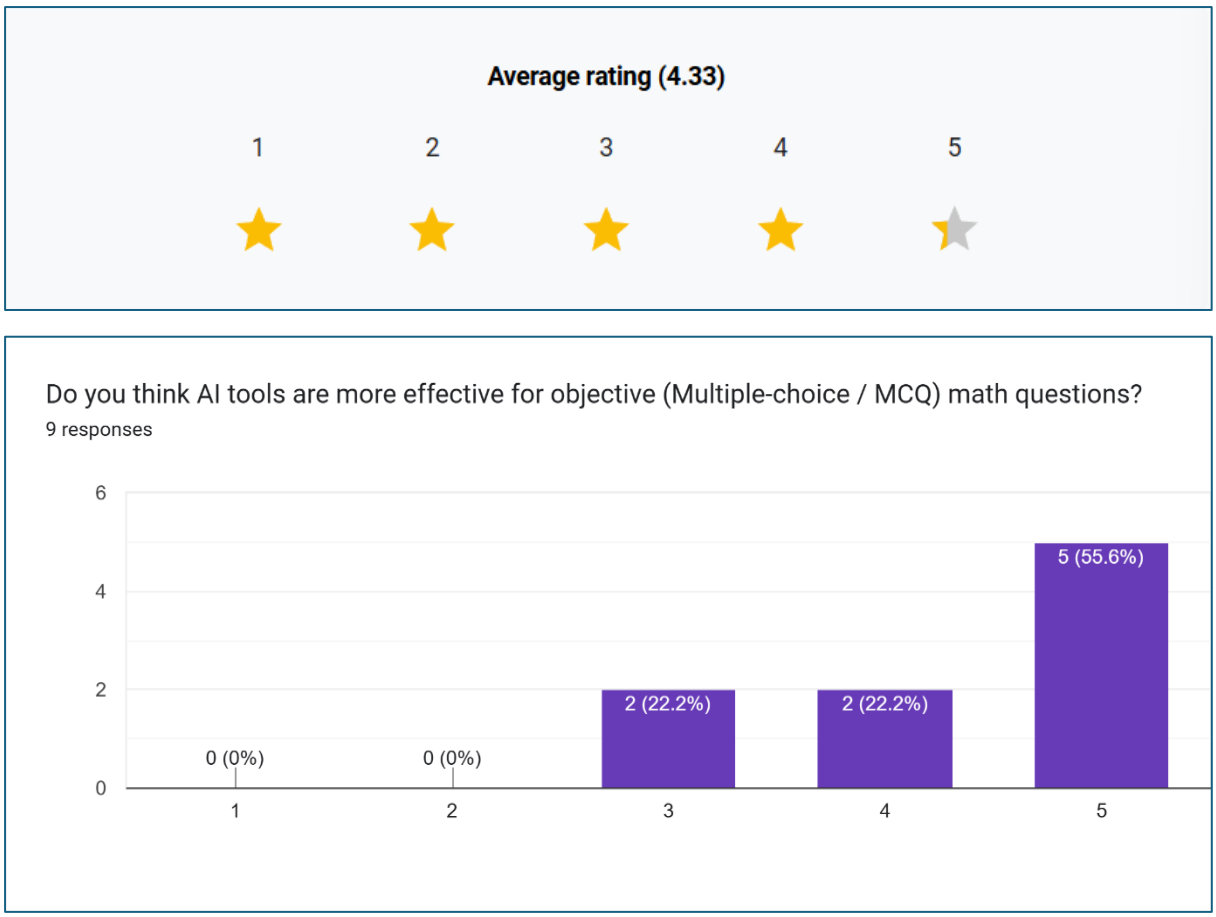


Figure 6.1.2.8: Rating AI tools efficiency for MCQ

Figure 6.1.2.8 presents ratings on the effectiveness of AI tools for MCQ questions. The highest percentage is 55.6% for 5-star rating, 22.2% for both 3-star and 4-star rating. Everyone thinks it is effective for MCQ, although it still got some error in AI answer generator.

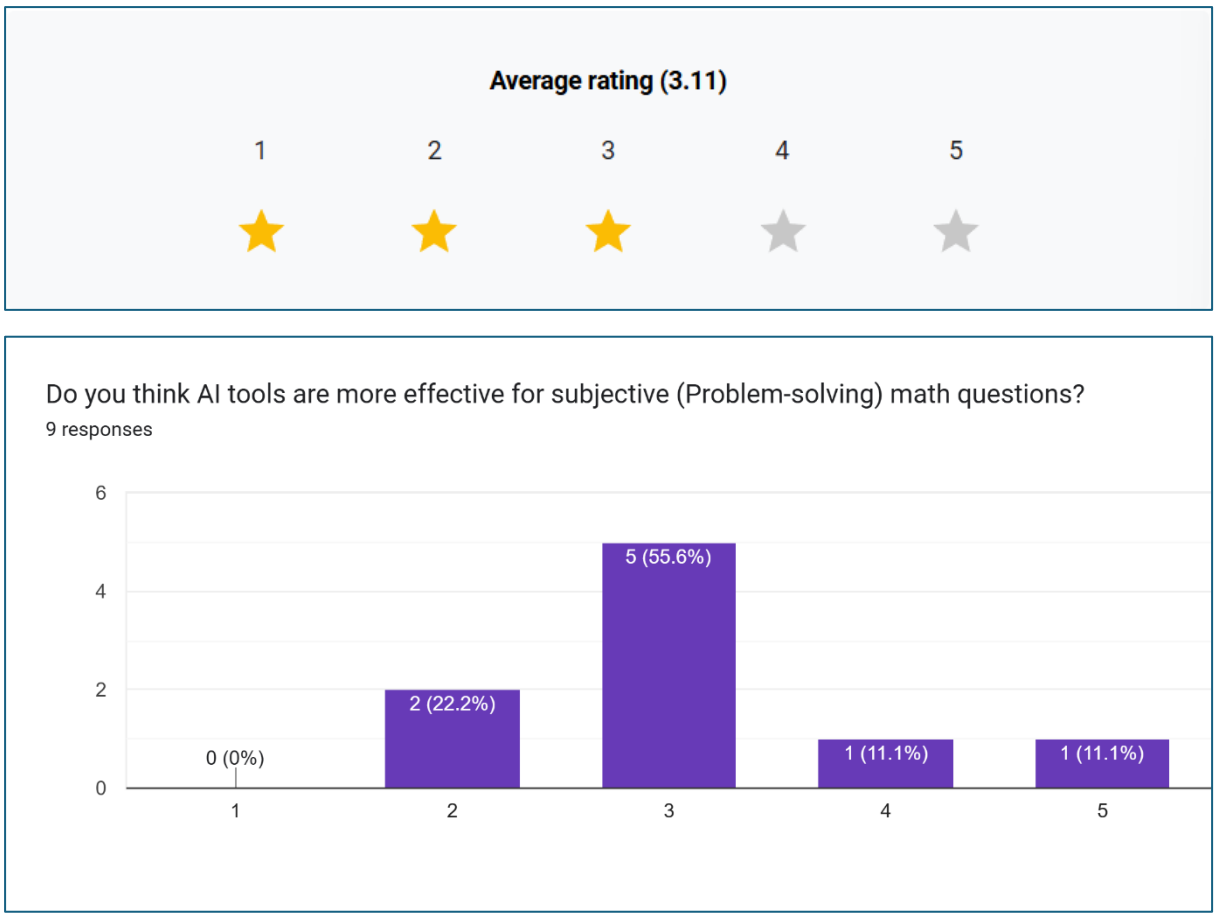


Figure 6.1.2.9: Rating AI tools efficiency for Subjective

Figure 6.1.2.9 presents rating on the effectiveness of AI tools for Subjective questions. The highest percentage is 55.6% for 3-star rating, 22.2% for 2-star rating, 11.1% for both 4-star and 5-star rating. These result shows that AI tools are not very effective for subjective question because of the AI's answer accuracy is low in subjective question.

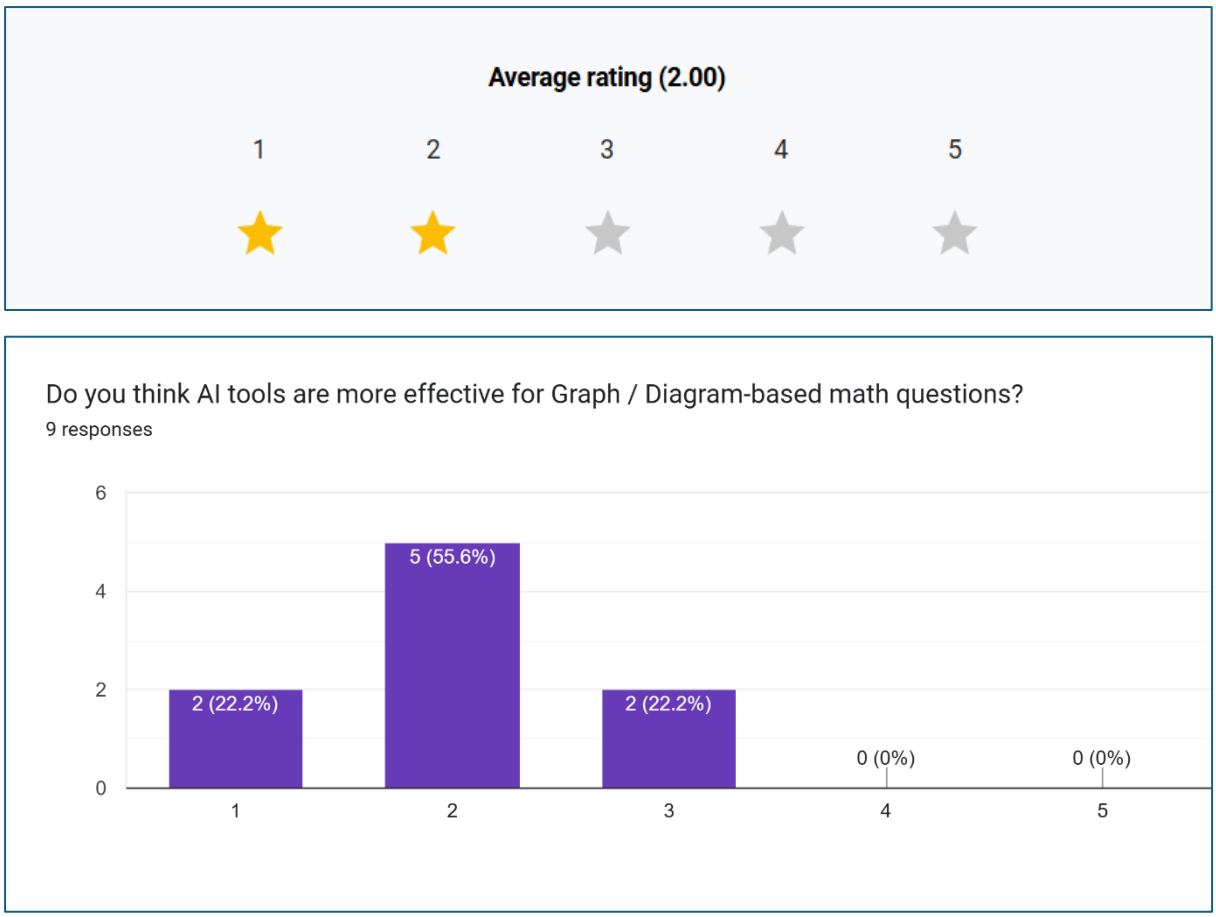


Figure 6.1.2.10: Rating AI tools efficiency for Subjective

Figure 6.1.2.10 presents rating on the effectiveness of AI tools for Graph/Diagram-based questions. The highest percentage is 55.6% for 2-star rating, 22.2% for both 1-star and 3-star rating. These result shows that AI tools are generally ineffective for Graph-based questions, as they often struggle to interpret and solve visual or diagram-related problems accurately.

Do you think AI tools are helpful in your math learning / teaching? Why or why not?

9 responses

Yes. AI can show the calculations step by step.

Yes, AI can show the calculations step by step.

https://www.savemyexams.com, has question bank and uses ai to mark subjective questions... just a matter of time before ai marking graph and long questions becoming a thing.

Useful but struggles with diagrams

AI gives breakdown steps I never thought of

Helpful for quick checking

AI improves marking efficiency but misinterprets graphs

Yes. It can give me answer directly without wasting too much time to think

Yes, although calculations are often wrong but it gives us the method

Figure 6.1.2.11: Open Questions from the Respondent

Figure 6.1.2.11 shows the open questions from the Respondent. The responses highlighted a range of suggestions, including:

- AI can show the calculations step by step.
- ai marking graph and long questions becoming a thing
- Useful but struggles with diagrams
- Helpful for quick checking
- AI gives breakdown steps I never thought of
- AI improves marking efficiency but misinterprets graphs
- give me answer directly without wasting too much time to think
- although calculations are often wrong, but it gives us the method
- These results suggest that students and teacher generally find AI tools helpful for understanding methods and checking calculations, especially for step-by-step guidance.

6.2 Functional Requirements

AI-Powered Problem Solving:

- The system shell provides AI-generated solutions for SPM math questions submitted by users.
- Users shall be able to take a photo or type a math question, and the AI shall return step-by-step solutions.
- The AI engine shall automatically detect the final answer produced during the solution process.
- The system shall retrieve the corresponding correct answer from the database and compare it with the AI-generated answer to validate accuracy.
- The system shall display both AI-generated answer and database's correct answer to the user for transparency.

Camera Function:

- The system shall allow the use of the device's camera, enabling users to take a photo and upload to the AI tutor.
- The system shall provide a scanning function to detect and extract the picture's answers from uploaded image. This allows the system to convert the extracted data into a readable format, making it easier to compare the student's answer with the correct answer stored in the database.

Text Arrangement Function:

- The AI-generated solution must follow a clear and standardized text format, such as step-by-step explanation are arranged in a top-to-bottom sequence.

SPM Math Question Practice:

- The system shall provide Form 4 & 5 practical questions, consisting of 16 chapters + 1 additional final practice section
- Each chapter shall have 3~5 questions in problem solving method.

6.3 Non-functional Requirements

Operational Requirements

- The system shall operate on Android mobile devices
- The application shall function both online and offline, where basic features such as chapter viewing and practice questions remain accessible offline, while AI-related features require an active internet connection.
- The application shall support common Android devices specifically, require Android API 34 or above
- The system shall integrate with the device's camera and storage permissions to enable scanning and image uploading.
- The system shall remain functional across various screen sizes and resolutions in different android devices.

Performance Requirements

- The AI-powered solver shall generate solutions in 5~10 seconds for each question.
- All primary interfaces (Home, Chapter, AI Tutor) shall be loaded in 3 seconds under a normal device performance condition.
- The scanned image shall be done in 5~10 seconds, uploaded to AI tutor shall be done in 3~5 seconds.

Security Requirements

- The system shall not require users to log in or provide personal data, ensuring that is no personally identifiable information (PII) is collected.
- All communications between Smart Math and AI server shall be secured using HTTPS protocols.

Reliability Requirements

- AI services shall maintain a minimum uptime of 95% during weekdays, ensuring consistent availability of the AI tutor.

Hand drawing Mathematics

- The system shell provides hand drawing function for user drawing mathematics question.

6.4 Modelling

6.4.1 Use Case Diagram

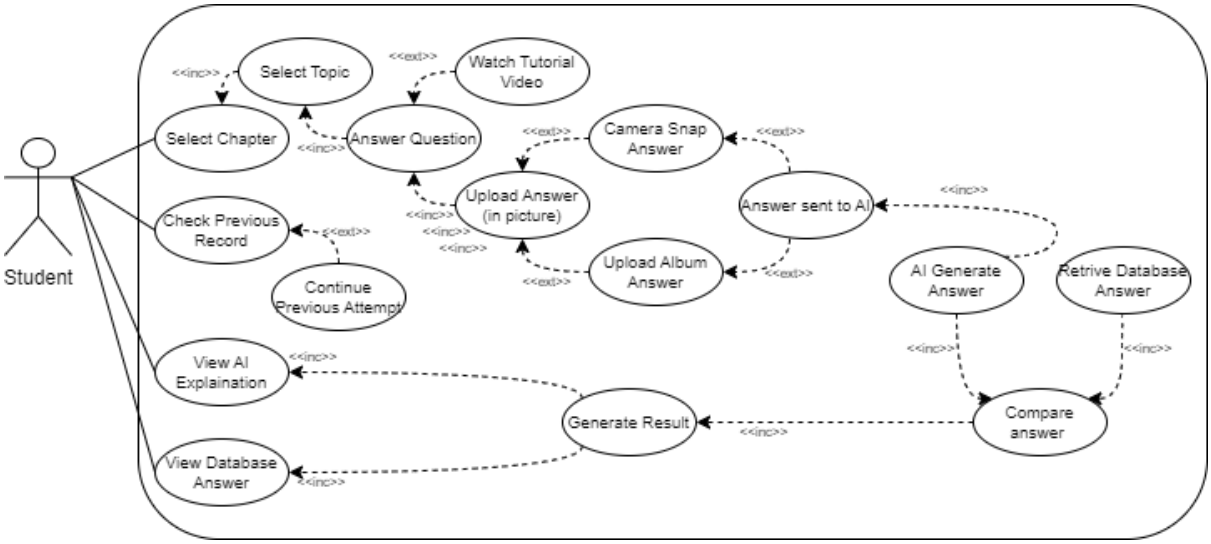


Figure 6.4.1 Use Case diagram

6.4.2 Use Case Description

Use Case Name	Select Chapter
Pre-condition	User opened the app
Post condition	User select the topic in the chapter
Flow of events 1. User opened the application 2. User select the chapter he wish to improve	

Use Case Name	Select Topic
Pre-condition	User had selected chapter
Post condition	User start answers the question
Flow of events 1. User select the chapter 2. User select the topic in the specific chapter (if any)	
Alternative Flow of Event 1. If the chapter has no topics, system directly proceeds to question selection.	

Use Case Name	Answer Question
Pre-condition	User had selected the topic in the chapter
Post condition	Users watch the tutorial video/upload the answer
Flow of events 1. User opened the application 2. User select the chapter and topic he wishes to improve 3. User select the question in the topic 4. User started to answer on paper	
Alternative Flow of Event 1. User skips the question and system loads another question.	

Use Case Name	Watch Tutorial Video
Pre-condition	User has selected a topic or completed answering a question.
Post condition	User gains understanding from the tutorial video.
Flow of events 1. System displays the tutorial video option on the question or result screen. 2. User selects "Watch Tutorial Video." 3. System loads the tutorial video player. 4. User watches the video to understand the solution or related concept.	
Alternative Flow of Event 1. User pauses, rewinds, or fast-forwards the video. 2. User closes the video before finishing.	

Use Case Name	Upload Answer
Pre-condition	Users answer the question
Post condition	Answer send to AI
Flow of events 1. User upload the answer using camera on the phone after the written down the answer on paper.	
Alternative Flow of Event 1. User can upload the answer from the album if they have stored the photo in album	
Exceptions 1. Image upload fails.	

Use Case Name	Camera Snap Answer
Pre-condition	User is answering a question and is ready to upload the handwritten answer.
Post condition	Answer send to AI
Flow of events 1. User selects the option to take a photo using the in-app camera. 2. System activates the device camera. 3. User aligns the written answer within the frame and takes a photo. 4. System displays a preview of the captured image. 5. User confirms the captured image to proceed.	
Alternative Flow of Event 1. User retakes the photo if the preview image is unclear. 2. User cancels and returns to the question screen without uploading.	
Exceptions 1. Camera permission not granted. 2. Camera fails to open. 3. Photo capture fails or image is corrupted.	

Use Case Name	Answer sent to AI
Pre-condition	User upload and check the answer they upload, ensure it is readable
Post condition	AI will generate the answer for the question
Flow of events 1. Users take photo or upload from gallery, ensure the photo is readable 2. The answer is sent to AI for checking	

Use Case Name	AI Generate Answer
Pre-condition	AI received the LaTeX format answer, analyse and find error
Post condition	AI will double check the answer with the answer in database.
Flow of events 1. AI receives the LaTeX format answer, then analyse it, find If there is a mistake exists. 2. When a mistake exists, AI will highlight the mistake part and show the correct path to user.	

Use Case Name	Retrieve Database Answer
Pre-condition	AI requires reference answer for verification.
Post condition	System check its generate answer is correct
Flow of events 1. System get the answer from database	
Exceptions 1. Database answer not found. 2. Database connection error.	

Use Case Name	Compare Answer
Pre-condition	System have generated its answer and retrieve database answer
Post condition	System generates the result to explain the answer
Flow of events 1. AI will compare the answer its generate to ensure it procedure is correct, this is used to prevent AI generate wrong answer.	
Alternative Flow of Event 1. AI finds mismatched results and adjusts its explanation.	

Use Case Name	Generate Result
Pre-condition	AI have compared the answer between its generate and database
Post condition	User available to look on the AI explanation and database answer.
Flow of events 1. When AI ensure its answer is correct, It will reply to the explanation and answer to user	
Exceptions 1. Result generation fails.	

Use Case Name	View AI Explanation
Pre-condition	System have retrieved the result to user interface
Post condition	-
Flow of events 1. AI have done it conclusion, and the reply have sent back to user interface	
Exceptions 1. Explanation fails to load.	

Use Case Name	View Database Answer
Pre-condition	System have retrieved the result to user interface
Post condition	
Flow of events 1. User allow to look into the answer in database for another solution for the same question	
Exceptions 1. Database answer display fails.	

Use Case Name	Check Previous Record
Pre-condition	User have done any question before, no matter completed or uncompleted
Post condition	-
Flow of events 1. User can look on his previous record to become a revision for himself	
Exceptions 1. Previous records fail to load.	

Use Case Name	Continue Previous Attempt
Pre-condition	User have an uncompleted record
Post condition	User can attempt to the question
Flow of events 1. If user having an uncompleted question, user are allowed to continue attempt for the question	
Exceptions 1. Record cannot be restored due to system error.	

System Design

7.1 Architectural Design

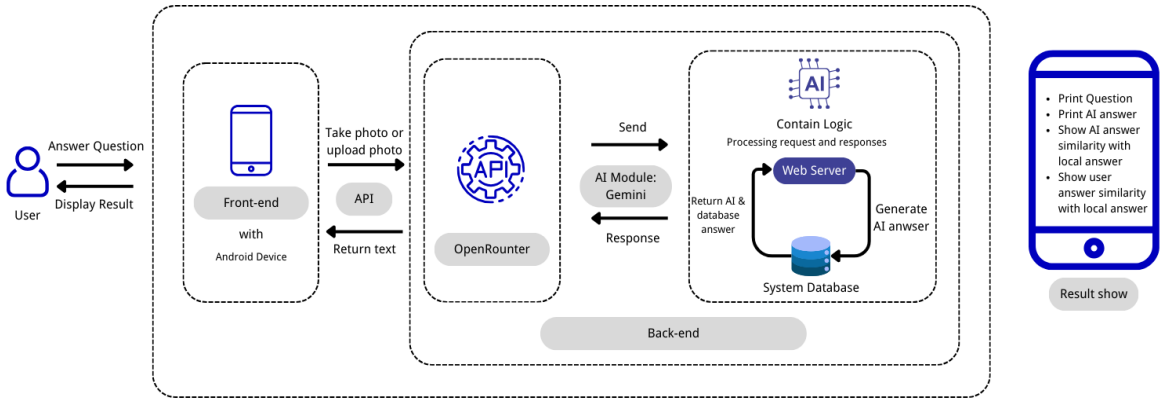


Figure 7.1.1 Architectural Design

The Smart Math application adopts a client-server architecture designed to support image-based math problem scanning, step-by-step solution assistance and structured chapter-based practice. Unlike AI-generated question systems, all math questions in Smart Math are manually curated and stored in the system, ensuring accuracy and alignment with the Form 4 & 5 syllabus.

User Interaction Layer

- The user answers math questions using take a photo or upload gallery photo of a question.
- The app displays the result, correctness, hints and steps.

Front-End Layer (Android Application)

- The front-end is developed using Android Studio
- Functions:
  - Allow user to answer chapter-based questions
  - Capture or upload a math image
  - Display validated result and solutions returned by the server

API

- Receive the answer in picture from mobile client
- Performs logic processing:
  - Match text with stored math questions
  - Validate the user's answer using stored solutions
  - Generate explanations steps using rule-based logic or simple AI processing
- Sends a structured response back to the mobile application

Database Layer

- Store all hardcoded chapter-based questions
- Stores correct answer, solution steps
- Stores user history (chapter progress)
- Support matching or validating answer

7.2 Database Design

7.2.1 Entity Relationships Diagram (ERD)

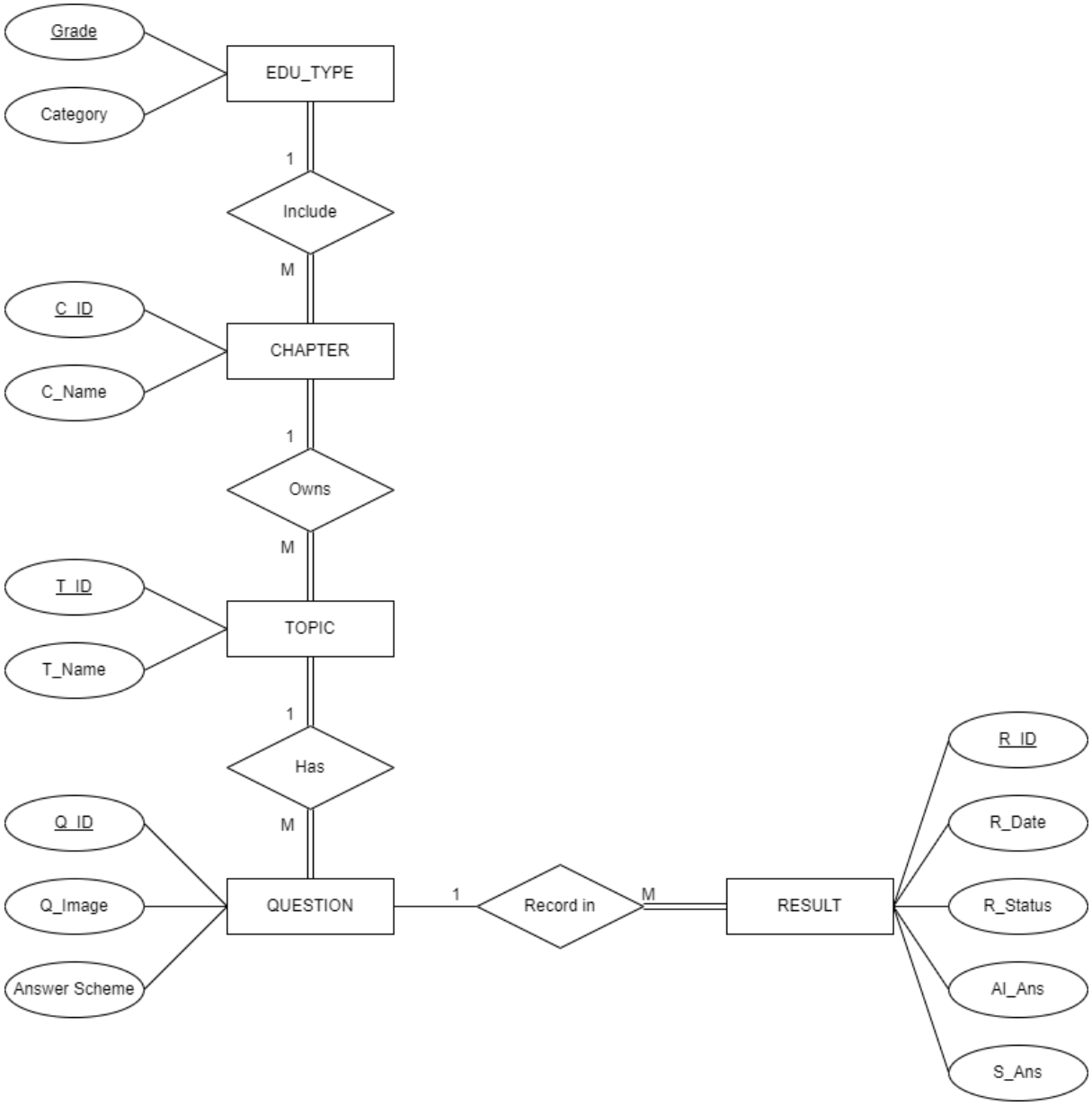


Figure 7.2.1.1 Entity Relationships Diagram (ERD)

7.2.2 Relational Schema

EDU_TYPE {Grade, Category)
CHAPTER {C_ID, C_Name, Grade(fk)}
TOPIC {T_ID, T_Name, C_ID(fk)}
QUESTION {Q_ID, Q_Image, Answer Scheme, T_ID(fk)}
RESULT {R_ID, R_Date, R_Status, AI_Ans, S_Ans, Q_ID(fk)}

7.2.3 Data Definition Language

```
CREATE DATABASE MATHS;

USE MATHS;

CREATE TABLE EDUTYPE
(Grade VARCHAR(255) PRIMARY KEY,
 Category VARCHAR(255));

CREATE TABLE CHAPTER
(C_Id VARCHAR(255) PRIMARY,
 C_Name VARCHAR(255),
 Level VARCHAR(255),
 CONSTRAINT fk_CHAPTER_EDUTYPE FOREIGN KEY (Grade) REFERENCES EDUTYPE
(Grade));

CREATE TABLE TOPIC
(T_Id VARCHAR(255) PRIMARY KEY,
 T_Name VARCHAR(255),
 C_Id VARCHAR(255),
 CONSTRAINT fk_TOPIC_CHAPTER FOREIGN KEY (C_Id) REFERENCES CHAPTER (C_Id));

CREATE TABLE QUESTION
(Q_Id VARCHAR(255) PRIMARY KEY,
 Q_Image VARCHAR(255),
 Q_AnswerScheme VARCHAR(255),
 T_Id VARCHAR(255),
 CONSTRAINT fk_QUESTION_TOPIC FOREIGN KEY (T_Id) REFERENCES TOPIC (T_Id));

CREATE TABLE RESULT
(R_Id VARCHAR(255) PRIMARY KEY,
 R_Date DATE,
 R_Status BOOLEAN COMMENT'Ongoing, Completed',
 AI_Answer VARCHAR(255),
 U_Answer VARCHAR(255),
 Q_Id VARCHAR(255),
 CONSTRAINT fk_RESULT_QUESTION FOREIGN KEY (Q_Id) REFERENCES QUESTION (Q_Id));
```

7.2.4 Data Dictionary

maths

chapter

Column	Type	Null	Default	Links to	Comments	Media type
C_Id (Primary)	varchar(255)	No				
C_Name	varchar(255)	Yes	NULL			
Grade	varchar(255)	Yes	NULL	edutype -> Grade		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	C_Id	0	A	No	
fk_CHAPTER_EDUTYPE	BTREE	No	No	Grade	0	A	Yes	

edutype

Column	Type	Null	Default	Links to	Comments	Media type
Grade (Primary)	varchar(255)	No				
Category	varchar(255)	Yes	NULL			

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	Grade	0	A	No	

question

Column	Type	Null	Default	Links to	Comments	Media type
Q_Id (Primary)	varchar(255)	No				
Q_Image	varchar(255)	Yes	NULL			
Q_AnswerScheme	varchar(255)	Yes	NULL			
T_Id	varchar(255)	Yes	NULL	topic -> T_Id		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	Q_Id	0	A	No	
fk_QUESTION_TOPIC	BTREE	No	No	T_Id	0	A	Yes	

result

Column	Type	Null	Default	Links to	Comments	Media type
R_Id (Primary)	varchar(255)	No				
R_Date	date	Yes	NULL			
R_Status	tinyint(1)	Yes	NULL		Ongoing, Completed	
AI_Answer	varchar(255)	Yes	NULL			
U_Answer	varchar(255)	Yes	NULL			
Q_Id	varchar(255)	Yes	NULL	question -> Q_Id		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	R_Id	0	A	No	
fk_RESULT_QUESTION	BTREE	No	No	Q_Id	0	A	Yes	

topic

Column	Type	Null	Default	Links to	Comments	Media type
T_Id (Primary)	varchar(255)	No				
T_Name	varchar(255)	Yes	NULL			
C_Id	varchar(255)	Yes	NULL	chapter -> C_Id		

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	T_Id	0	A	No	
fk_TOPIC_CHAPTER	BTREE	No	No	C_Id	0	A	Yes	

7.3 Prototype Design

Figure 7.3.1 Content for Smart Math

Figure 7.3.1 shows the main content of Smart Math. This page displays the application logo and a brief description of the app.

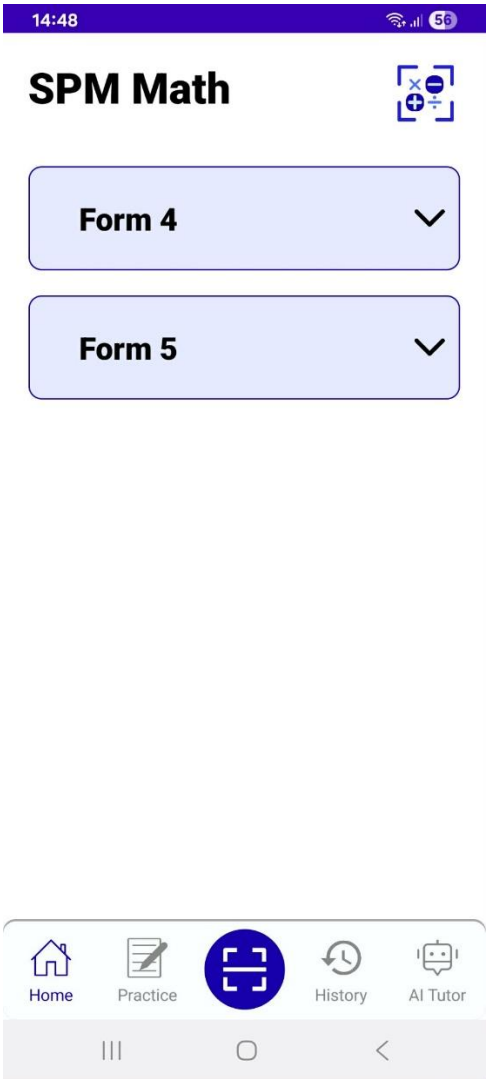


Figure 7.3.2 Home Page

Figure 7.3.2 shows the Home Page. User can choose “Form 4” or “Form 5” to practice the mathematics question. The application does not require any user login or registration.

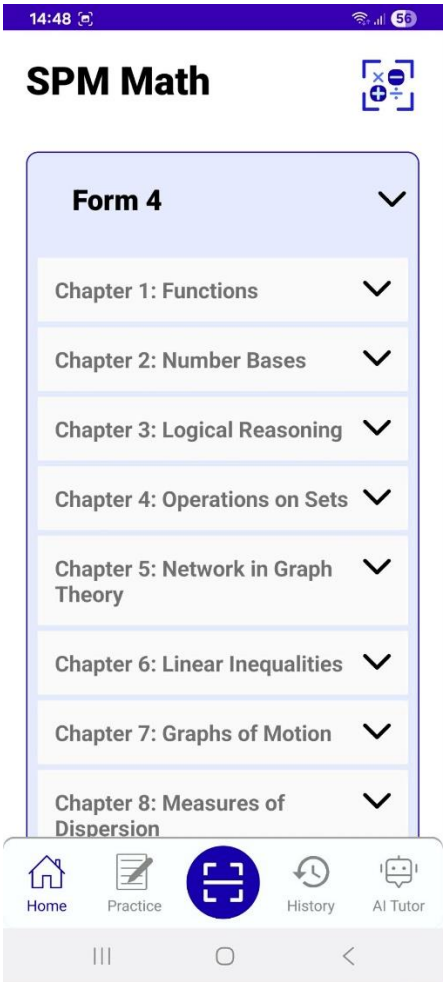


Figure 7.3.3 Home Page (After Click Form 4)

Figure 7.3.3 show the Chapter selection in Form 4, where users can browse and select a specific chapter

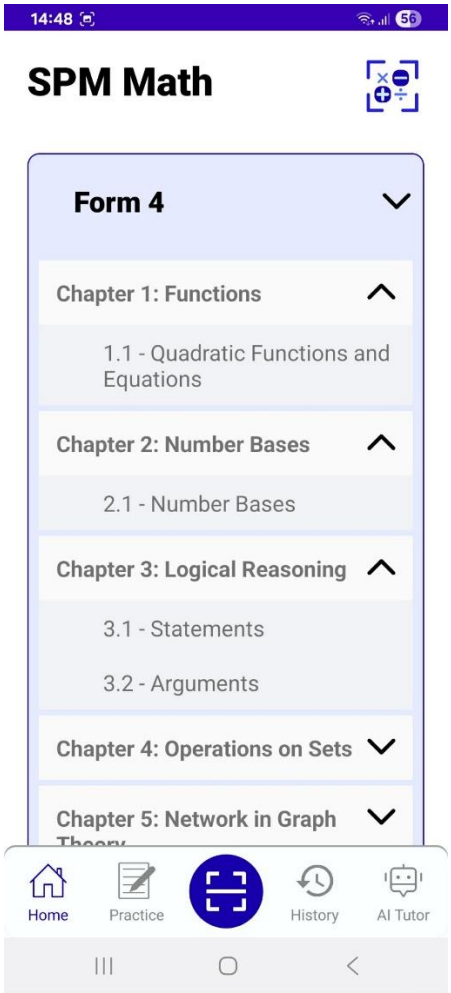


Figure 7.3.4 Home Page (After Click Chapter 1)

Figure 7.3.4 show the topic selection in Chapter 1. User can choose any topic in Chapter1, each topic at least has 1 question.

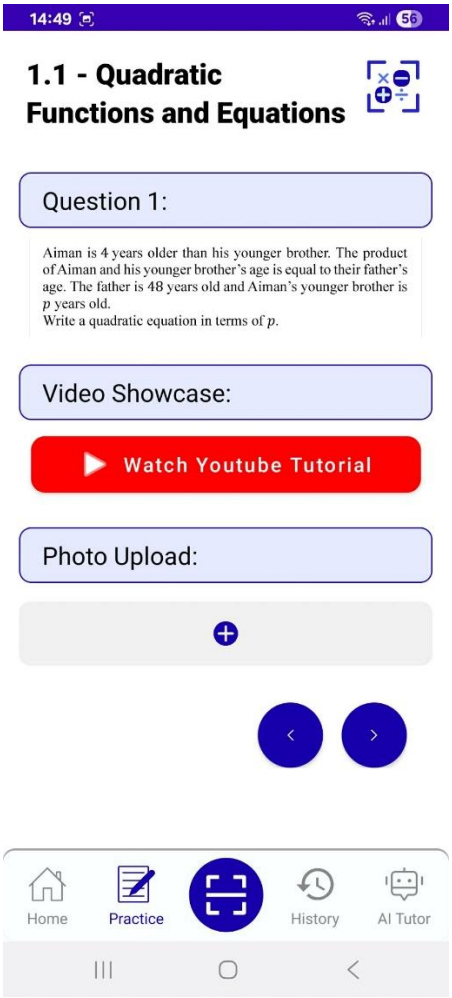


Figure 7.3.5 Practice Page (After Click Topic 1.1)

Figure 7.3.5 shows the topic 1.1 question, video showcase and photo upload in Practice Page. User can view the question image with zoom-in and zoom-out the image. User can click the video directly within the application or open the video on YouTube.

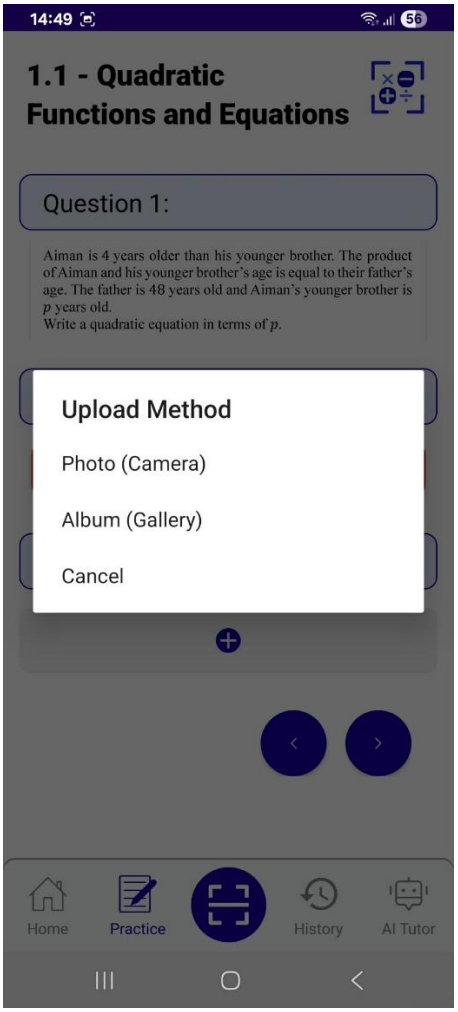


Figure 7.3.6 Practice Page (After Click Upload Button)

Figure 7.3.6 shows the upload options for uploading an answer. User can choose to upload from the photo, album or google album.

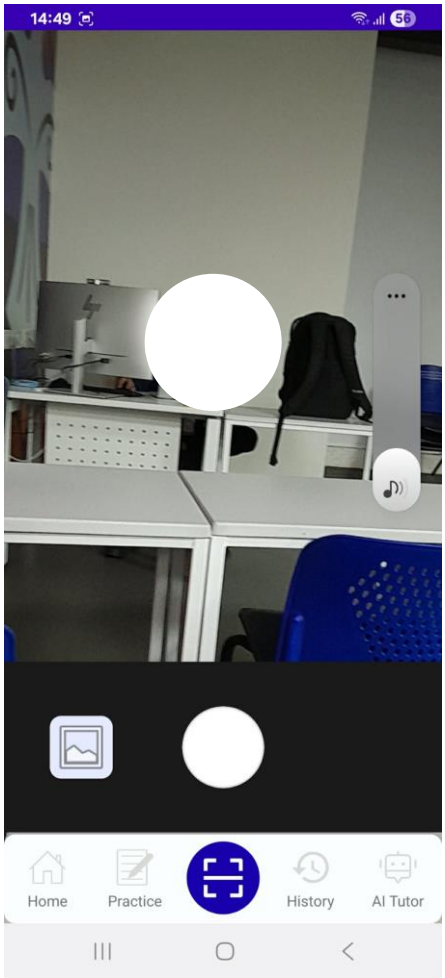


Figure 7.3.7 Scan Page (After Click photo)

Figure 7.3.7 shows the camera function in Scan Page. User can click the scan button (blue color button) to scan the mathematics question. User also can choose the photo button (left) or drawing math button (right) to select and upload the image to AI tutor.

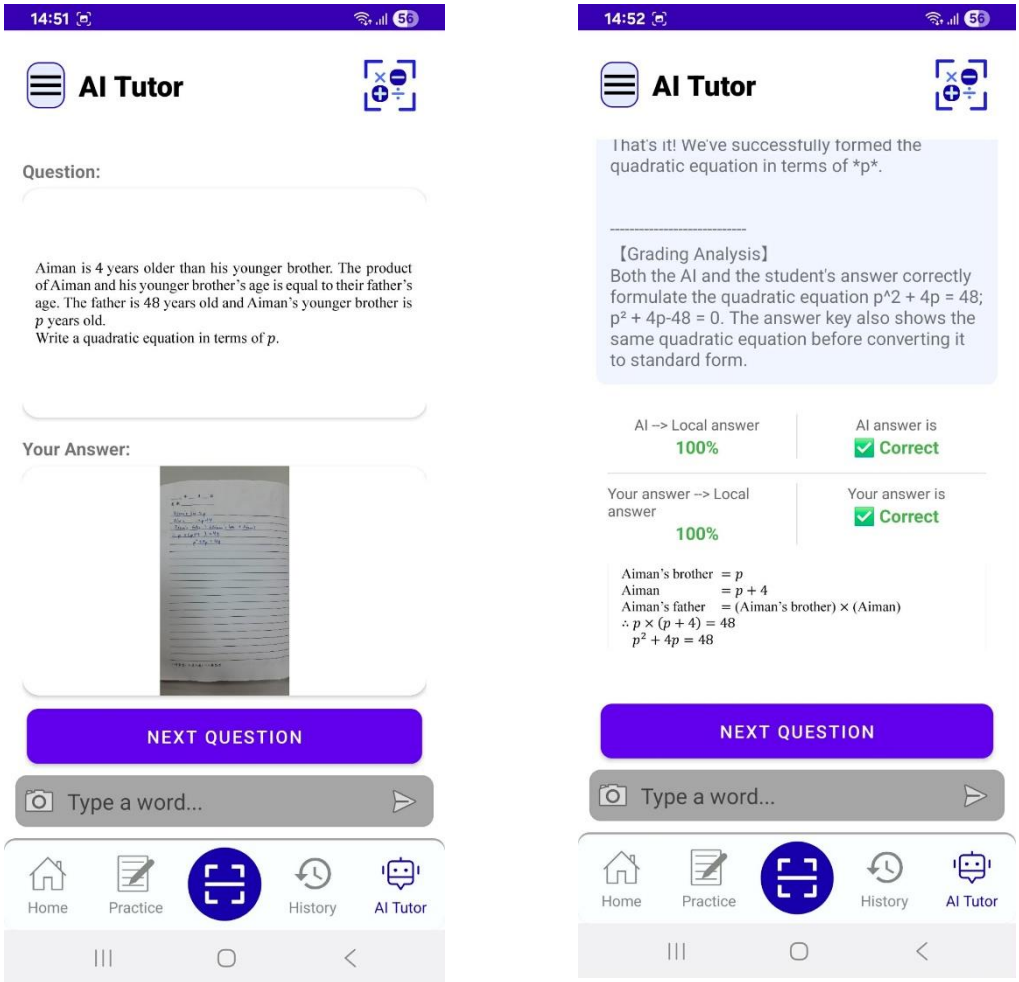


Figure 7.3.8 AI Tutor Page (After Uploading)

Figure 7.3.8 shows the chat with AI tutor. User can view the answer by AI tutor, correct answer (database answer) and the similarity of AI answer compare with database answer. User also can choose to upload a new photo or chatting with AI tutor. There also has the similarity check for user, which can compare the AI answer to Local answer and User answer to local answer. The below part is the local answer.

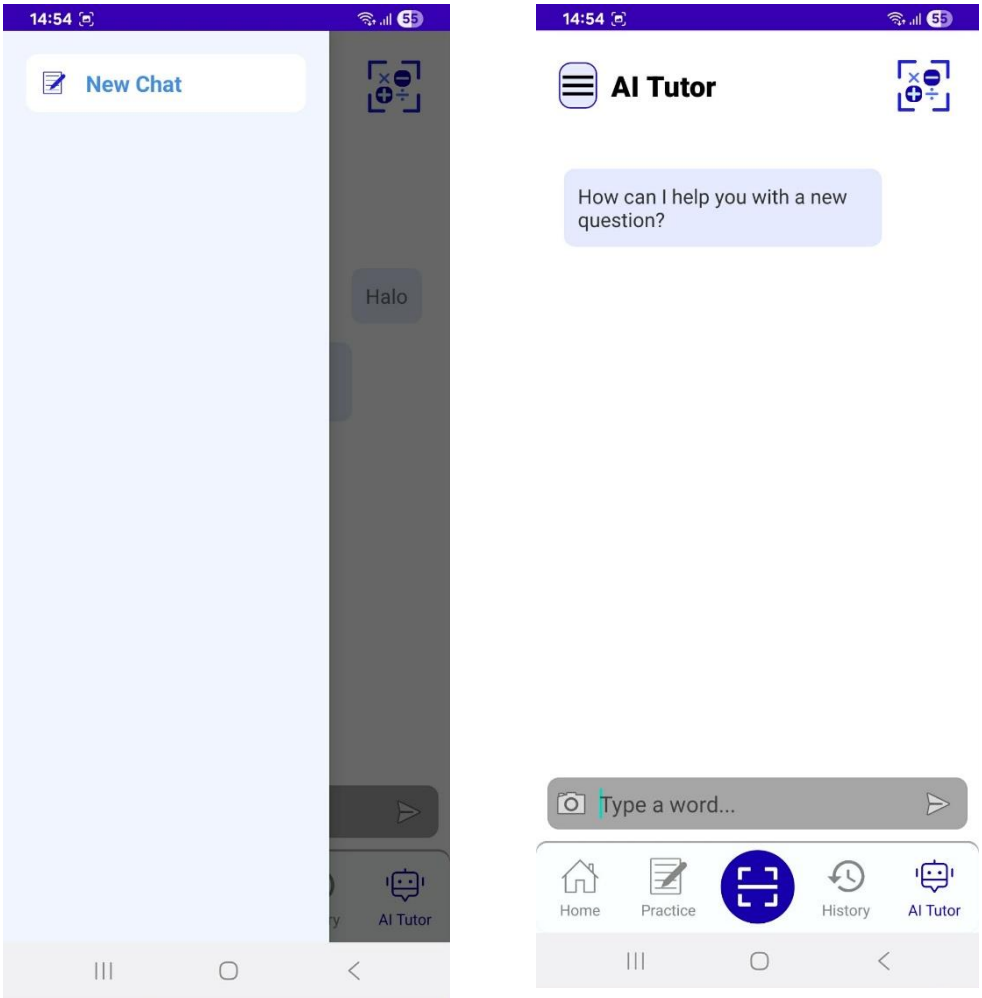


Figure 7.3.9 AI Tutor Page (After Click the menu bar)

Figure 7.3.9 shows the menu list in the AI tutor Page. User can create a new chat.

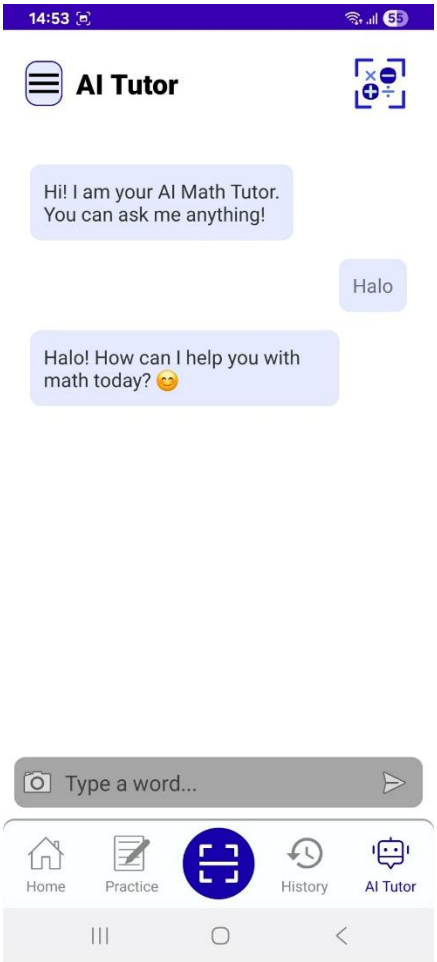


Figure 7.3.10 AI Tutor Page (use chat to asking question)

Figure 7.3.10 shows the chat-based method for asking questions. No accurate answer is shown because the question does not exist in the database

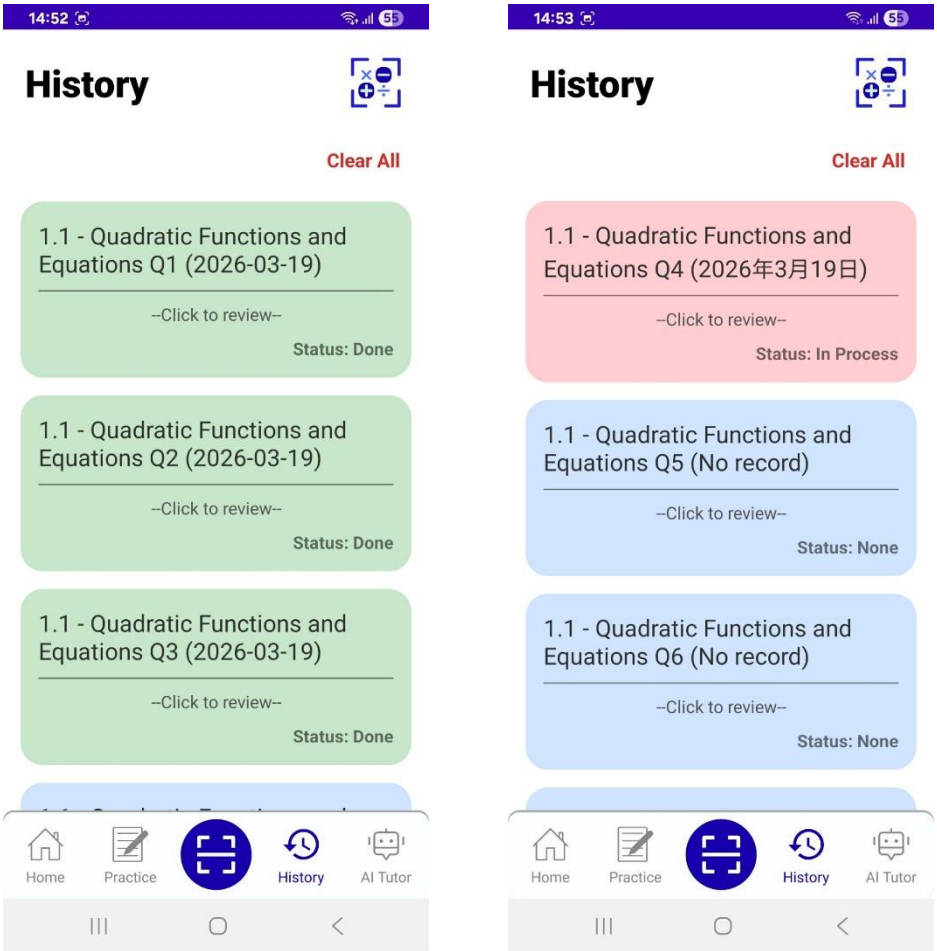


Figure 7.3.11 History Page (use chat to asking question)

Figure 7.3.11 shows the History Page, which displays a list of previously attempted questions and their statuses, such as “Done,” “In Process,” or “None” (not attempted before). Users can review these items and continue attempting unanswered or incomplete questions.

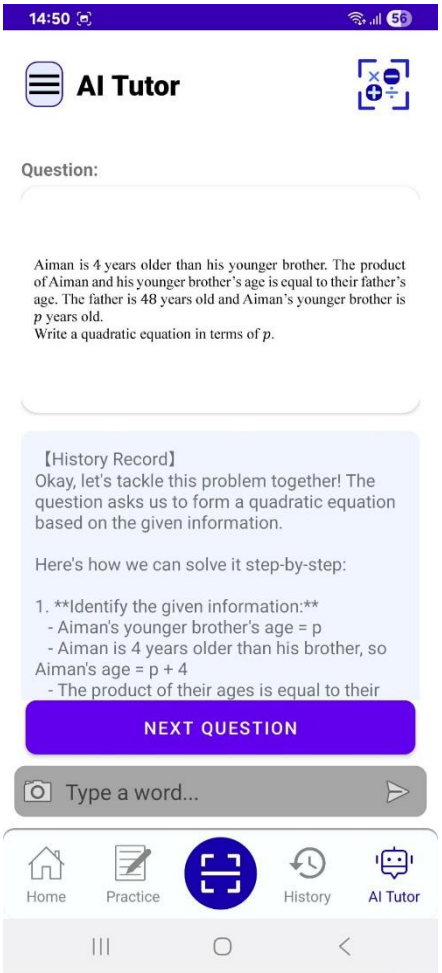


Figure 7.3.12 History Page to AI Tutor (Click the green bar)

Figure 7.3.12 show the history function after user click the “green bar”. It will show the question, AI step-by-step answer and local answer

System Deployment

8.1 Selected AI Chatbot

After a review and comparison with several AI Chatbot, Gemini 2.0 Flash from Open Router has been chosen as the AI tutor of the application. Gemini is an AI Chatbot develop and offered by Google. Gemini has been chosen due to these advantages:

- Higher accuracy: Gemini provides a higher accuracy on answering the maths question.
- Quick response: Gemini has provide a quicker response than other AI Chatbot

Figure 8.1.1 Logo of OpenRouter and Gemini AI

Discussion and Results

9.1 Achievements

Throughout the application development, below are the achievements measure up through the project:

- Successfully integrated AI Chat with SQLite data verification.
- Developed a user-friendly mobile interface for Form 4/5 syllabus
- Integrated a Review function for students to track their progress.

9.2 Limitations

Though the development process, there still remain areas uncovered based on limitations.

- **Client Server Database**: The failure of implementing client-server database have cause the question data, answer data and user data store locally, leads to a larger application size.

9.4 Future Enhancements

To improve the long-term capabilities and value of the Smart Math application, several enhancements are planned for future development. These improvement aims to upgrade the functionality, strengthen AI reasoning and provide a more adaptive learning for SPM students.

1. **AI-Generated Question Bank (Dynamic Question Generation)**

   The current Smart Math system stores all mathematics question manually using a local database. While this method works for the prototype stage, it becomes inefficient and difficult to maintain when the SPM syllabus changes or when new chapters and question variations are required.

   To address this limitation, future version of Smart Math will integrate AI-generated question capabilities. The system will dynamically produce SPM style mathematics question by:

   - Modifying numerical values
   - Generating different scenarios
   - Adjust difficulty levels
   - Producing multiple variations of the same concept

   But this feature also faced the challenge of verification mechanisms. Therefore, system must be ensured that the AI-generated solution can be automated validation.

2. **Enhanced Handwriting Recognition (Advanced OCR Model)**

   The current OCR engine can detect printed mathematical text but may struggle with handwritten or low-quality images. Future plans include:

   - Integrating a stronger math-focused OCR model (but the cost is expensive)
   - Improving symbol recognition (roots, fractions indices, inequalities)
   - Supporting full handwritten working steps for student input

3. **Personalized Learning and Progress Analytics**

   A student learning analytics module will be added to provide insights such as:

   - Weak chapters and topics (the question is not enough for students to practice or learning)
   - Accuracy trend across weeks
   - AI-guided recommendations on which chapter to revise

4. **Cloud Database Migration (Scalability Upgrade)**

   The current system uses SQLite for local storage. A future version may migrate to cloud-based solutions such as Firebase or Supabase to support:

   - Synchronised progress across devices
   - Long-term storage of student performance
   - Deployment to large groups of students nationwide

References

Agile Alliance. (n.d.). *Agile 101.* Retrieved October 5, 2025, from https://www.agilealliance.org/agile101

Canva. (2024). *About Canva*. Retrieved from https://www.canva.com/about/

ElSherief, M., Alqahtani, M., & AlGhamdi, A. (2023). *Drivers and consequences of ChatGPT use in higher education: A conceptual framework and research agenda.* Education and Information Technologies, 28, 14581–14599. https://pmc.ncbi.nlm.nih.gov/articles/PMC10670526/

Google Play Store. (2024). *User reviews for Homework Helper* [Mobile application]. LeoStudio Global Ltd. Retrieved from https://play.google.com/store/apps/details?id=homework.helper

Herold, B. (2024). *AI gets math wrong sometimes: How teachers deal with its shortcomings.* Education Week. https://www.edweek.org/teaching-learning/ai-gets-math-wrong-sometimes-how-teachers-deal-with-its-shortcomings/2024/09

Kissflow. (2024). *Rapid Application Development Platform* [Web article]. Retrieved from https://kissflow.com/application-development/rad/rapid-application-development-for-business-transformation/

LeoStudio Global Ltd. (n.d.). *Homework Helper* [Mobile application]. Google Play Store.
Available at https://play.google.com/store/apps/details?id=homework.helper

Lim, S. M., Tan, H. K., & Chong, W. Y. (2021). *Challenges in mathematics learning among Malaysian secondary school students: A case study approach*. Malaysian Journal of Education, 46(2), 55–63. https://ejournal.ukm.my/jpend/article/view/51102

Mathpix, Inc. (n.d.). *Mathpix Snip* [Mobile application]. Google Play Store.
Available at https://play.google.com/store/apps/details?

Meet Android Studio. (n.d.). Android Developers. Retrieved March 6, 2024, from https://developer.android.com/studio/intro?hl=zh-cn

Lembaga Peperiksaan Malaysia. (2021). *Format Pentaksiran Sijil Pelajaran Malaysia (SPM): Matematik 1449 (Kemaskini 7 Jun 2021)*. Ministry of Education Malaysia.
https://lp.moe.gov.my/images/bahan/spm/2021/BukuFormatSPM2021/FORMAT%20SPM%202021%201449%20MATEMATIK%20updated%207%20jun.pdf

OpenAI. (2024). *ChatGPT API usage limits and image generation quota*. Retrieved from https://platform.openai.com/docs/guides/rate-limits

OpenRouter. (2024). *OpenRouter API Documentation*. Retrieved from https://openrouter.ai/

Orji, E. Z., Haydar, A., Erşan, İ., & Mwambe, O. O. (2023). *Advancing OCR accuracy in image-to-LaTeX conversion — A critical and creative exploration*. *Applied Sciences, 13*(22), 12503. https://doi.org/10.3390/app132212503

Sajja, P., & Jonita, R. (2024). *AI-based tools in mathematics education: A systematic review of characteristics, applications, and evaluation methods*. *International Research Journal on Advanced Engineering Hub*. https://doi.org/10.47392/IRJAEH.2024.0268

Sheikh Rahim, M. (2025, April 29). *84,025 gagal Matematik*. *Utusan Malaysia*. Retrieved from https://www.utusan.com.my/nasional/2025/04/84025-gagal-matematik/

Yeo, J. B. W. (2010). *Students' difficulties in mathematics problem-solving: What do they say?* Procedia - Social and Behavioral Sciences, 8, 142–151. https://doi.org/10.1016/j.sbspro.2010.12.020

Appendix

10.1 Progress Form

DIT2175 IT Project
Progress Report Form

SUNWAY COLLEGE

Date : 7/10/2025 Week No. : 8

Updates from the previous meeting

No previous meeting.

Items discussed this meeting

- Revision of proposal: add references in each chapter.
- Discussion on the scope of the writing project plan and Requirement Gathering
- Discussion on conducting of interview
- Selection of database for the project.
- Discussion on system design scope

Works for the coming meeting

- IT Project Proposal (Chapter 5-6)
- interviews with SPM teachers (at least 3 people) and surveys for SPM students (at least 30 people).
- Research and write the requirement of RAD and Agile methodology needed
- Design an EERD diagram
- Design mathematic questions (Form **4** Chapter 1-8)
- Start to build an API connection and try to get the answer from AI.
- Research a server database for data storage and retrieval.

Supervisor's Signature

.....
(Ms. Seetha Letchumy M.Belaidan)

Student's Signature

.....
(Koh Pei Yang)

Student's Signature

.....
(Chin Kai Yang)

DIT2175 IT Project
Progress Report Form

SUNWAY COLLEGE

Date : 27/10/2025 Week No. : 11

Updates from the previous meeting

- Proposal: Completed fact-finding (minimum 30 SPM students & 3 teachers), refine the Functional Requirements and System Design.

Items discussed this meeting

- Design of mathematics questions, including marking rubric and question logic.
- Verification of OCR results — allowing users to choose whether to upload the converted text form, since the OCR-generated text may contain inaccuracies after image conversion.

Works for the coming meeting

- Implement the OCR technique and integrate it into the application.
- Build a return gateway for OCR results to allow users to decide whether to upload the converted text.

Supervisor's Signature

.....
(Mr Kevin Foo Kai Pin)

Student's Signature

.....
(Koh Pei Yang)

Student's Signature

.....
(Chin Kai Yang)

DIT2175 IT Project
Progress Report Form

SUNWAY COLLEGE

Date : 26/11/2025 Week No. : 15

Updates from the previous meeting

- Complete the proposal chapter

Items discussed this meeting

- Reviewed and discussed the overall report structure to ensure the content is logical and reasonable.
- Confirmed the ERD Diagram, Use Case Diagram, and System Architecture flow.
- Discussed improvements and additional points needed for the proposal

Works for the coming meeting

- Finalize and refine the DIT2166 IT Project Proposal based on feedback.
- Begin implementing OCR and LaTeX functions in Android Studio.
- Start the testing process for the overall program to ensure proper functionality and performance.

Supervisor's Signature

.....
(Ms Seetha Letchumy M.Belaidan)

Student's Signature

.....
(Koh Pei Yang)

Student's Signature

.....
(Chin Kai Yang)

DIT2175II IT Project
Progress Report Form

SUNWAY COLLEGE

Date : 4/2/2026 Week No. : 2

Updates from the previous meeting

- Camera (scan) function, AI Tutor function complete
- Math Form 4 all chapter question and answer complete

Items discussed this meeting

- Application progress and improvement for the application
- Report writing and improvement

Works for the coming meeting

- Continue the application (add on similarity function, zoom picture function, ai prompt function)
- Continue the report (add on methodology, testing result and remark the use case)
- Testing application (find around 5 people to test the app)
- Check the report for similarity index

Supervisor's Signature

.....
(Ms Seetha Letchumy M.Belaidan)

Student's Signature

.....
(Koh Pei Yang)

Student's Signature

.....
(Chin Kai Yang)

DIT2175II IT Project
Progress Report Form

SUNWAY COLLEGE

Date : 18/3/2026 Week No. : 7

Updates from the previous meeting

- Basic function of application had already done.
- The improvement for the application has partially done.

Items discussed this meeting

- Missing several parts in report (need to add into before finalizing)

Works for the coming meeting

- Finalize the report and fulfil all the needed

Supervisor's Signature

.....
(Ms Seetha Letchumy M.Belaidan)

Student's Signature

.....
(Koh Pei Yang)

Student's Signature

.....
(Chin Kai Yang)

10.2 Survey Question

Smart Math Survey

This survey aims to collect opinions from both SPM students and teachers/tutors about how AI tools are used in mathematics learning and teaching.

In the first part, you will answer questions about your math practice habits.

* Indicates required question

What is your age? *

- ◯ Below 16
- ◯ 16 ~ 17 (SPM Level)
- ◯ 18 ~ 24 (College / University)
- ◯ Above 24

What is your role? *

- ◯ Student
- ◯ Teacher / Tutor
- ◯ Other: _____

How often you pratice math? *

- ◯ Daily
- ◯ Once a week
- ◯ Several times a month
- ◯ Rarely
- ◯ Never

Which type of math exercise do you prefer to practice? *

- ◯ Objective (Multiple-choice / MCQ)
- ◯ Subjective (Problem-solving)
- ◯ Graph / Diagram-based

Do you think **Objective (Multiple-choice / MCQ)** is hard? *

Diberi bahawa $5e - 4 = 16 - (e + 5)$, hitung nilai e.
Given that $5e - 4 = 16 - (e + 5)$, calculate the value of e.

A $\frac{6}{25}$

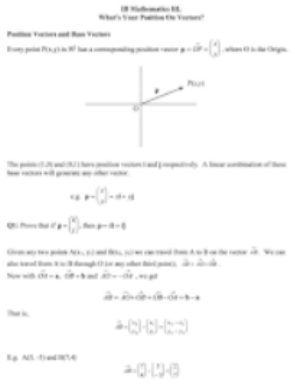
B $\frac{2}{5}$

C $\frac{5}{2}$

D $\frac{25}{6}$

1	2	3	4	5
☆	☆	☆	☆	☆

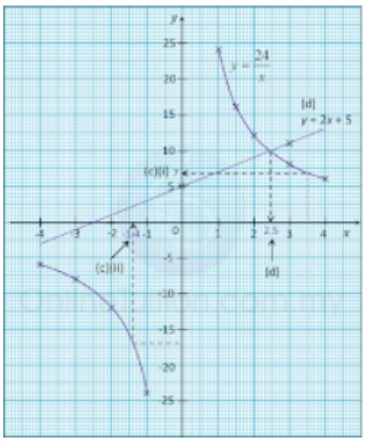
Do you think **subjective (Problem-solving)** is hard? *



1 2 3 4 5

☆ ☆ ☆ ☆ ☆

Do you think **Graph / Diagram-based** is hard? *



1 2 3 4 5

☆ ☆ ☆ ☆ ☆

Next

Clear form

Smart Math Survey

* Indicates required question

In the second part, you will answer questions about how you use AI in mathematics learning.

Which type of AI do you often use? *

- ◯ Chat-based AI (ChatGPT, DeepSeek)
- ◯ Math-solving AI (Photomath)
- ◯ Voice assistant AI (Siri, Google Assistant)
- ◯ None
- ◯ Other: _____

Do you think AI tools are more effective for **objective (Multiple-choice / MCQ)** math questions? *

1 2 3 4 5

☆ ☆ ☆ ☆ ☆

Do you think AI tools are more effective for **subjective (Problem-solving)** math questions? *

1 2 3 4 5

☆ ☆ ☆ ☆ ☆

Do you think AI tools are more effective for **Graph / Diagram-based** math questions? *

1 2 3 4 5

Do you think AI tools are helpful in your math learning / teaching? Why or why not? *

Your answer

Back Submit Clear form

Smart Math User Experience Survey

B I U

This survey is a follow-up to the **Smart Math study**, focusing on the user experience of AI tools in mathematics. It aims to collect further insights from both SPM students and teachers/tutors.

What is your name? *

Short-answer text

What is your role? *

- Student
- Teacher
- Other:

How would you rate your overall experience with our application? *

1 2 3 4 5

To what extent does this application help you in learning or teaching mathematics? *

1 2 3 4 5

Which specific feature did you find most engaging or helpful? *

Long-answer text

What improvements or additional features would you like to see in future updates? *

Long-answer text